

TECHNOLOGY



Designing Microsoft Azure Infrastructure Solutions: AZ:305

Design a Network Solution



Learning Objectives

By the end of this lesson, you will be able to:

- Analyze insights into virtual network concepts to provide private connectivity between Azure Virtual Machines and other Azure services
- Identify hub-and-spoke architecture for centralized control, streamlined communication, and simplified management
- Assess solutions for network security to safeguard against potential threats
- Develop strategies for automating network management to enhance efficiency, scalability, and responsiveness to dynamic changes



Learning Objectives

By the end of this lesson, you will be able to:

- Outline Azure Front Door load service to improve the performance and availability of web applications
- List the best practices of Azure CDN (Content Delivery Network) to optimize content delivery, reduce latency, and enhance web application



Recommend a Network Architecture

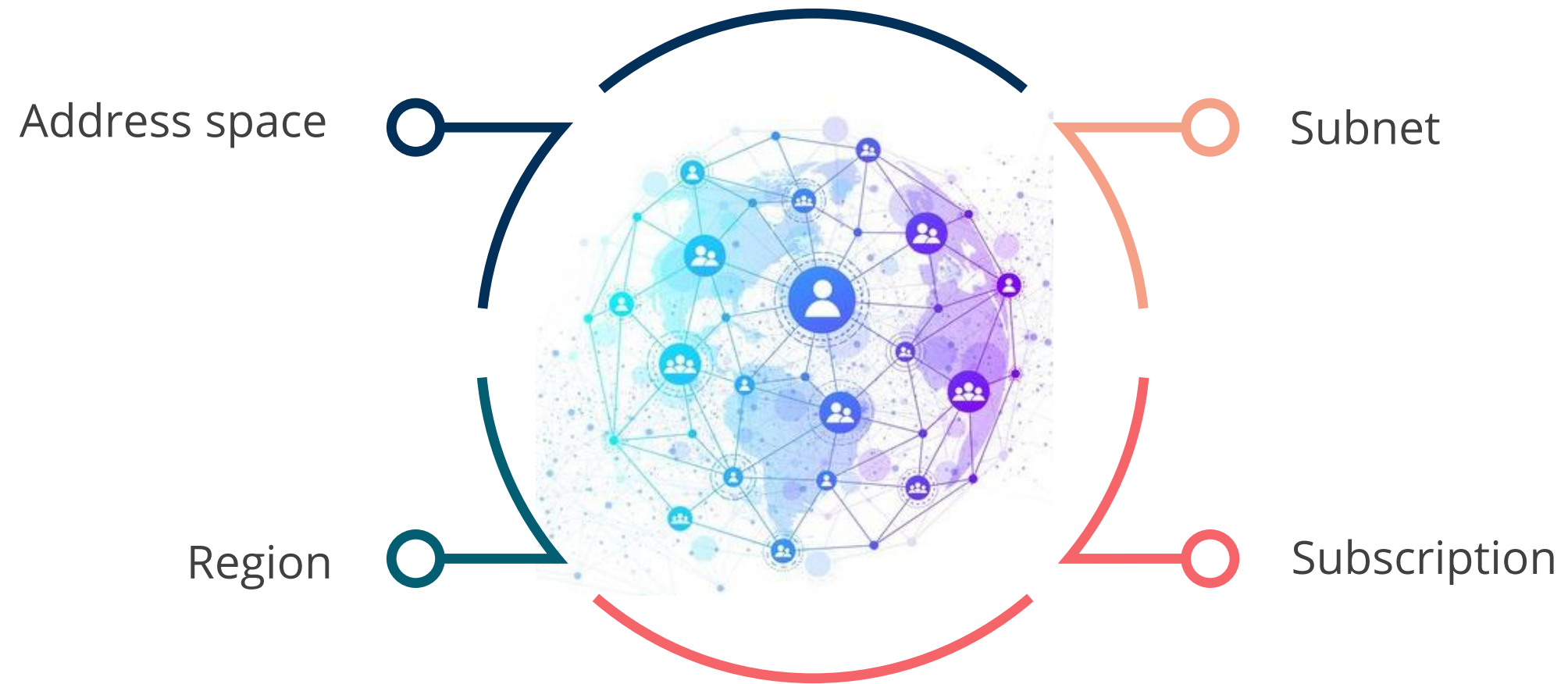
Virtual Network (VNet)

It is a logical representation of the network in the cloud.



It provides private connectivity between Azure Virtual Machines and other Azure services.

Virtual Network Concepts



Address Space

It refers to the range of IP addresses that can be used within a network.



When creating a VNet, the user must specify a custom private IP address space. By default, Azure assigns resources in a virtual network to a private IP from the address space that users assign.

Subnets

These are subdivisions of a larger network, allowing for better organization, security, and resource allocation.



Within a virtual network, subnets help segment the network into smaller, more manageable units.



Region

It refers to a geographic location where cloud resources, including virtual networks are hosted.



While creating VNet, a single region or location for deployment can be selected.

Subscription

It is associated with a specific account and is used to manage and govern the resources deployed within the virtual network.



Virtual networks are scoped to a subscription.



Hub-Spoke Network Topology

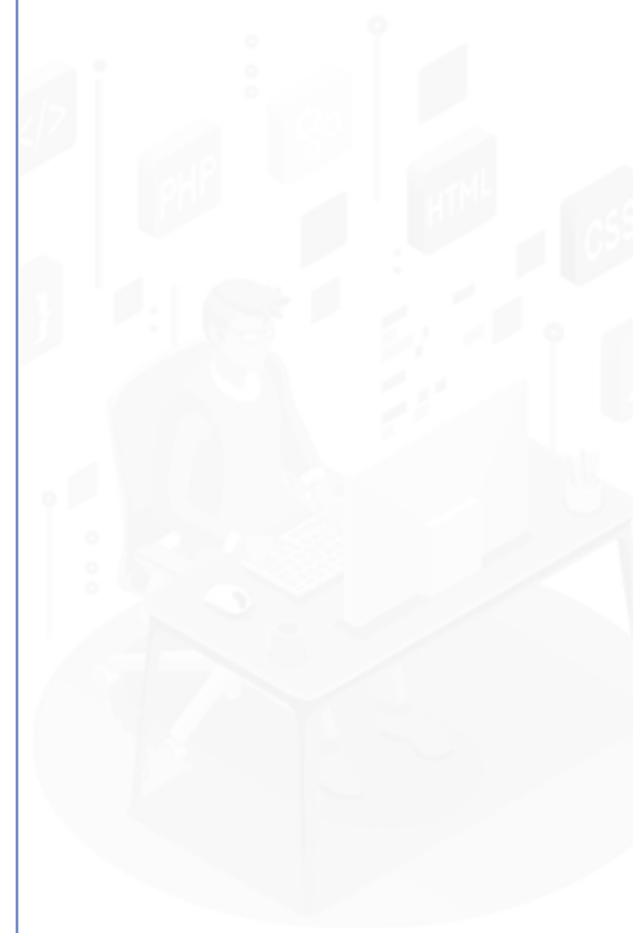
The hub virtual network acts as a central point of connectivity to many spoke virtual networks.

The hub can be used as the connectivity point for on-premises networks.

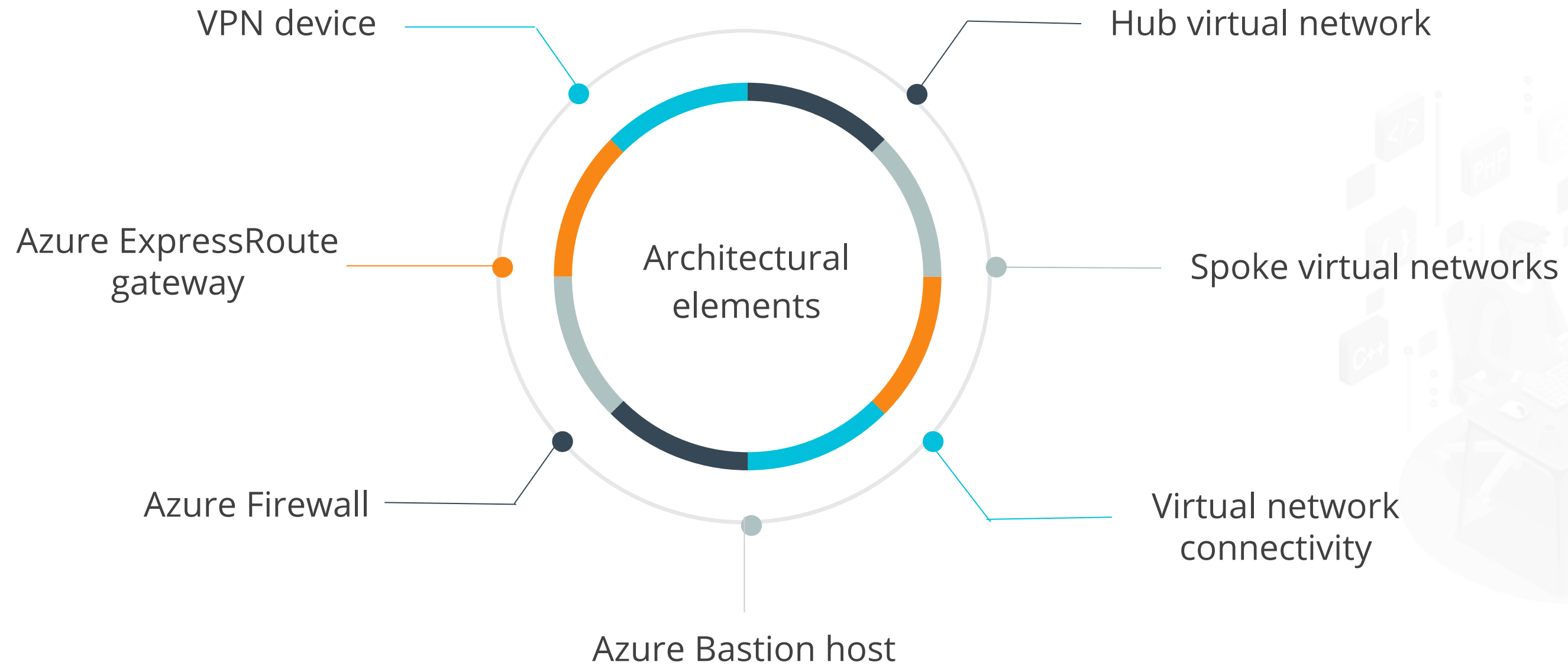
The spoke virtual networks peer with the hub and can be used to isolate workloads.



©Simplilearn. All rights reserved.



Hub-Spoke Architecture



Use Cases of Hub-Spoke Architecture



Workloads that require shared services are placed in the hub virtual network, while each environment is deployed to a spoke.



Workloads that require access to shared services but do not require connectivity to each other.



Enterprises that require central control and segregated management for the workloads in each spoke.

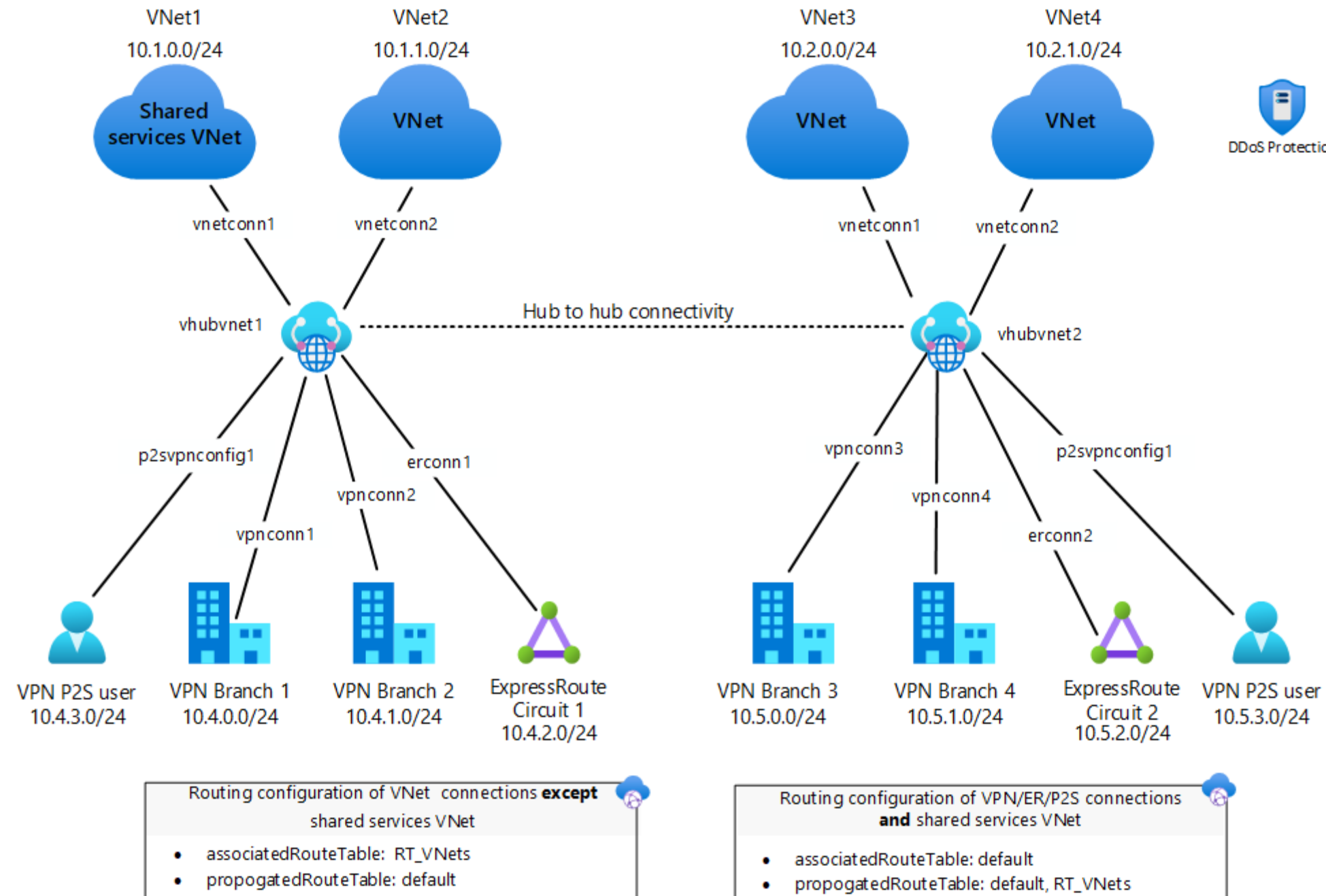
Hub-Spoke Network Topology with Virtual WAN

It offers an alternative to the reference hub-spoke network topology in Azure, enabling the implementation of a secure hybrid network.



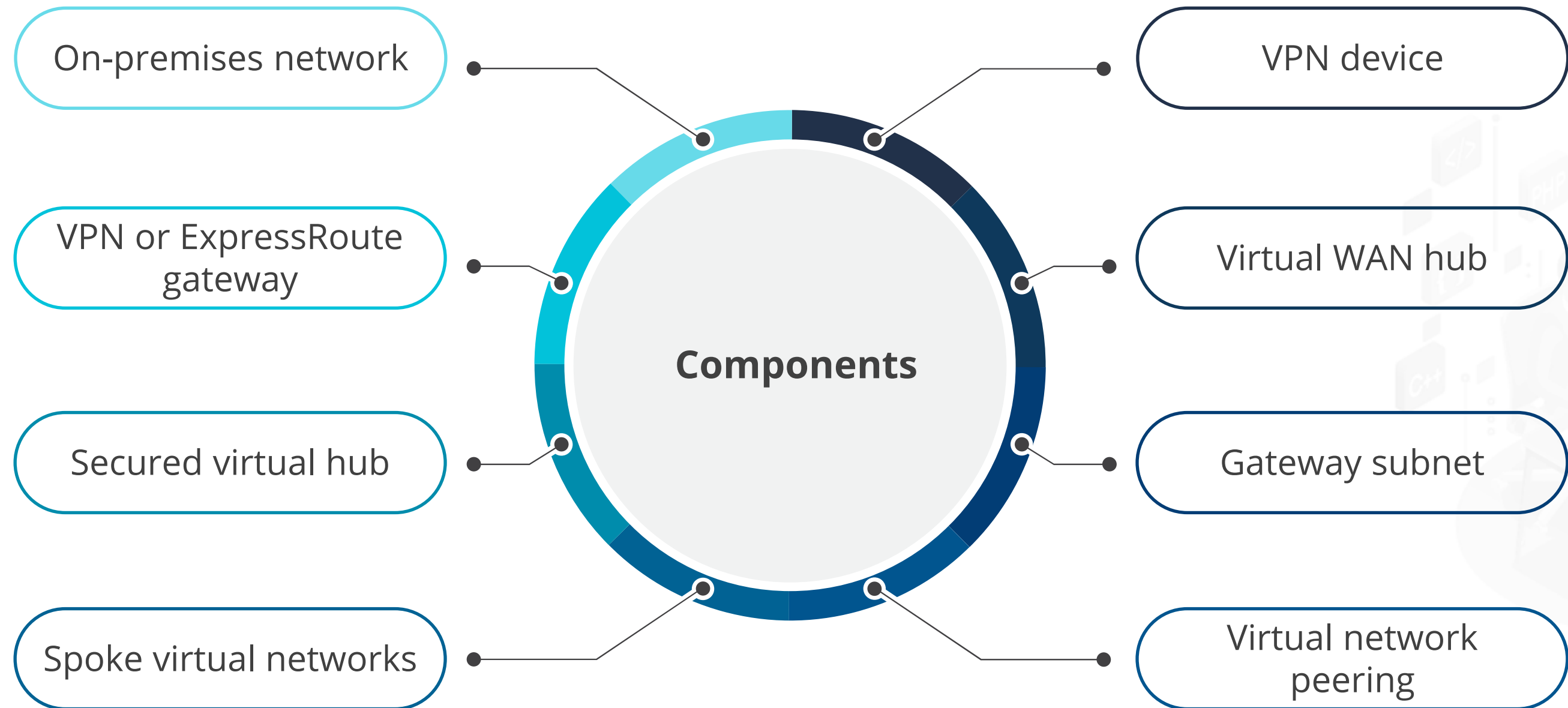
- Hub VNet is a connection point for several spoke VNets.
- Spoke VNets can be used to separate workloads via peering with the hub.
- Traffic travels between the on-premises data center(s) and the hub using an ExpressRoute or VPN gateway.

Hub-Spoke Network Topology with Virtual WAN



Source: <https://learn.microsoft.com/en-us/azure/architecture/networking/media/hub-spoke-vwan-architecture-001.png>

Hub-Spoke Network Topology with Virtual WAN: Components



Hub-Spoke Network Topology with Virtual WAN

Benefits

Less operational
overhead

Cost
savings

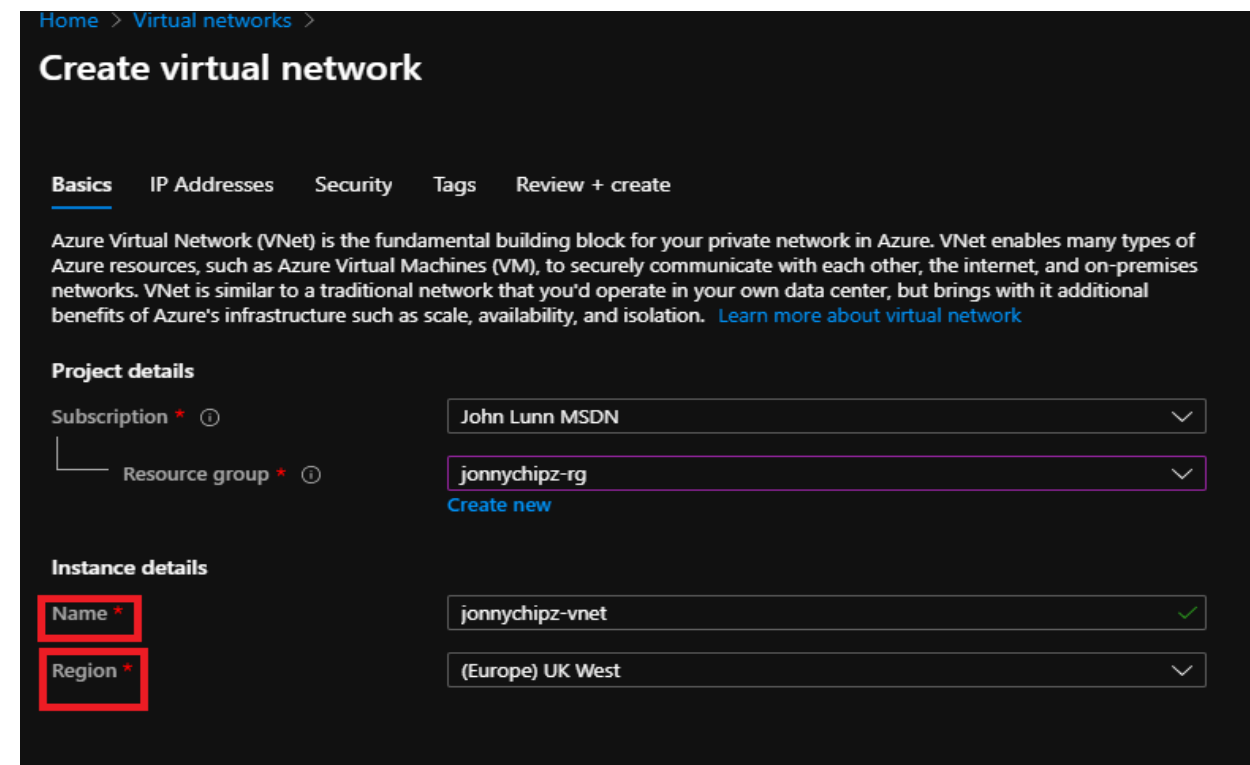
Improved
security

Separation of
concerns

Recommend a Solution for Network Provisioning

Name and Region

Each Azure resource has a unique name. Within a scope, the name must be unique, which varies depending on the resource type.



Home > Virtual networks >

Create virtual network

Basics IP Addresses Security Tags Review + create

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation. [Learn more about virtual network](#)

Project details

Subscription * ⓘ John Lunn MSDN

Resource group * ⓘ jonnychipz-rg
[Create new](#)

Instance details

Name * jonnychipz-vnet ✓

Region * (Europe) UK West

An Azure region produces all Azure resources. Only a virtual network in the same region can create a resource.

Segmentation

- The user can construct many virtual networks per subscription and location.
- Within each virtual network, the user can construct several subnets.
- Azure routes network traffic between all subnets in a virtual network.
- A virtual network can be split into two or more subnets up to the constraints.



Permissions and Policy



- Permissions are allowed for the management group, subscription, resource group, and individual resources.
- The user can create, assign, and manage policy definitions using Azure Policy.
- Azure Policy evaluates the resources of the user that are not compliant with the policy definitions.

Virtual Network Best Practices

-  Ensure non-overlapping address space
-  Ensure the subnet does not encompass the entire VNet address space
-  Configure a few larger VNets rather than many small ones
-  Secure VNets by assigning Network Security Groups to subnets



Recommend Solutions for Network Security

Network Security

It protects the communication of resources within and outside of a network.

Focus areas

It secures traffic flow between:

- Applications and the internet
- Among the applications
- Users and the application
- Applications and the internet limit exposure outside the network
- Applications and their tiers, environments, and services within a network



Filter Network Traffic

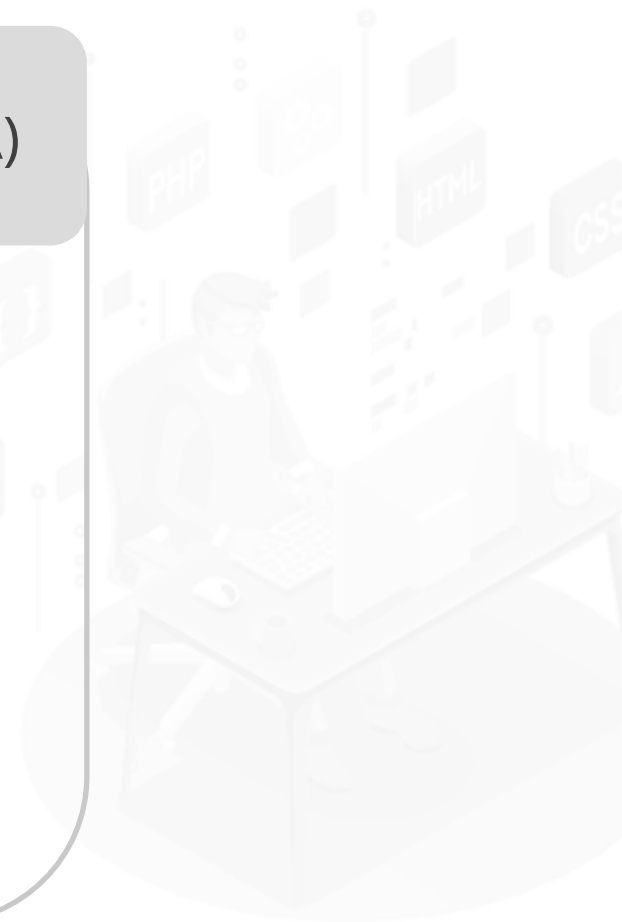
Network traffic can be filtered using:

Network Security Groups (NSG)

It contains a list of security rules that allow or deny inbound or outbound network traffic.

Network Virtual Appliances (NVA)

It is a VM that performs a network function, such as a firewall, WAN optimization, or other network functions.



Route Network Traffic

By default, Azure routes traffic between subnets, connected virtual networks, on-premise networks, and the Internet.

To override the default rules, the following can be used:

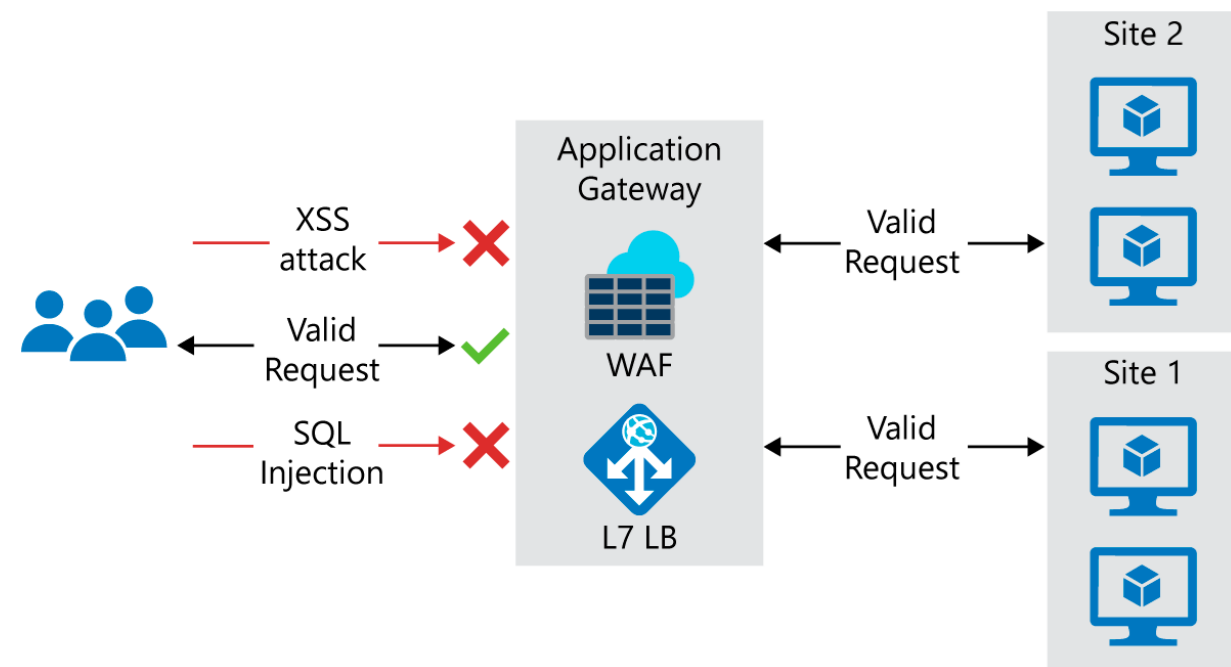
Route tables

A custom route table, containing routes that dictate traffic routing, can be created for each subnet.

Border Gateway Protocol (BGP) routes

An on-premise BGP route can be propagated to a VNet if a user links the VNet to an on-premise network, either through an Azure VPN Gateway or an ExpressRoute connection.

Internet Protection

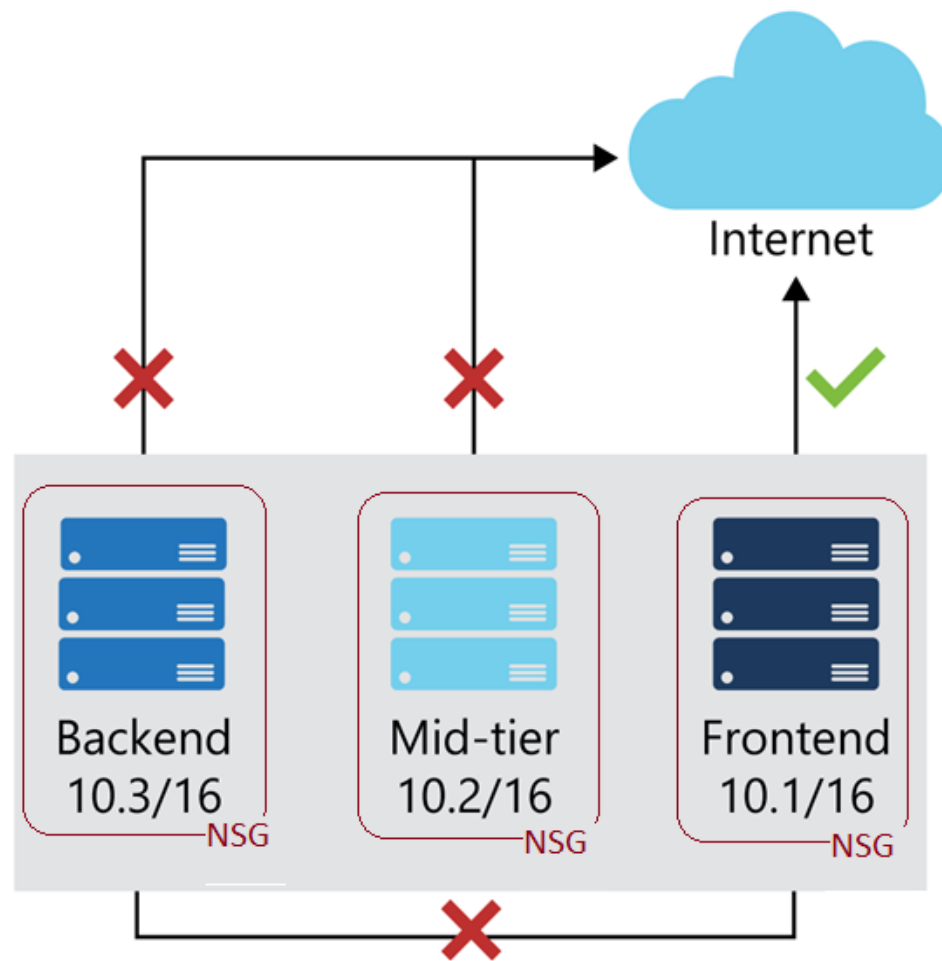


Different ways of restricting traffic from the internet:

- Identify resources that are allowing inbound network
- Use Azure Security Center
- Use an Application Gateway

Virtual Network Security

In an Azure virtual network, a user can use an Azure NSG to filter network traffic to and from Azure resources.



A user can use virtual network service endpoints to isolate services, enabling communication between virtual networks.

Network Security Group

Features

- Enables a user to filter network traffic to resources in a virtual network
- Contains a list of priorities that allow or deny inbound or outbound network traffic
- Can be associated with a subnet or a network interface
- Operates in layers 3 and 4, allowing communication between network interfaces
- Are used to isolate applications between environments, tiers, and services
- Locks down network communication between virtual machines



Network Security Group Rules

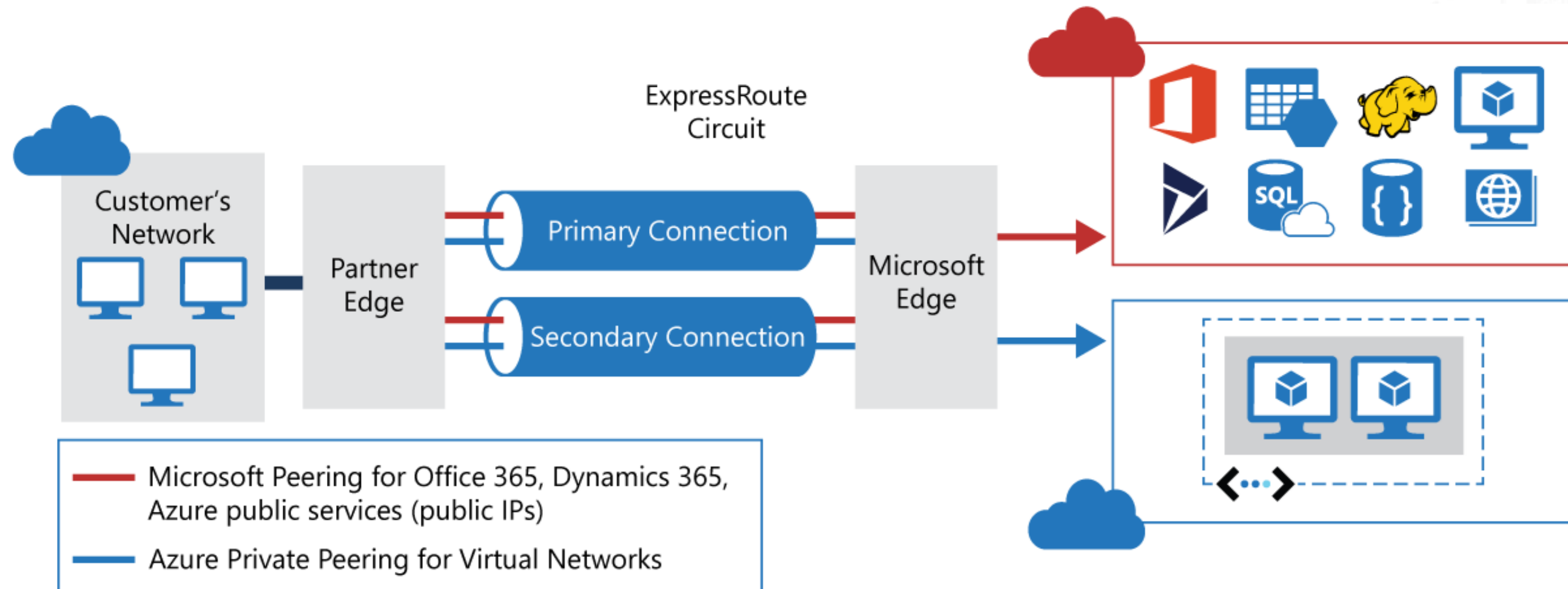
Security rules in NSGs enable a user to filter network traffic that can flow in and out of virtual network subnets and network interfaces.

VM1-nsg - Inbound security rules				
Network security group				
PRIORITY	NAME	PORT	PROTOCOL	ACTION
65000	AllowVnetInBound	Any	Any	✓ Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	✓ Allow
65500	DenyAllInBound	Any	Any	✗ Deny
VM1-nsg - Outbound security rules				
Network security group				
PRIORITY	NAME	PORT	PROTOCOL	ACTION
65000	AllowVnetOutBound	Any	Any	✓ Allow
65001	AllowInternetOutBound	Any	Any	✓ Allow
65500	DenyAllOutBound	Any	Any	✗ Deny

There are default security rules. A user cannot delete the default rules but can add other rules from priority 100 to 4096 with lower priority numbers.

Network Integration

VNet integration for an Azure service allows restricting access to the service exclusively to the associated virtual network infrastructure.



Recommend Solutions for Network Connectivity

Virtual Network Connectivity

Different ways to allow network connectivity between virtual networks are:



VNet peering



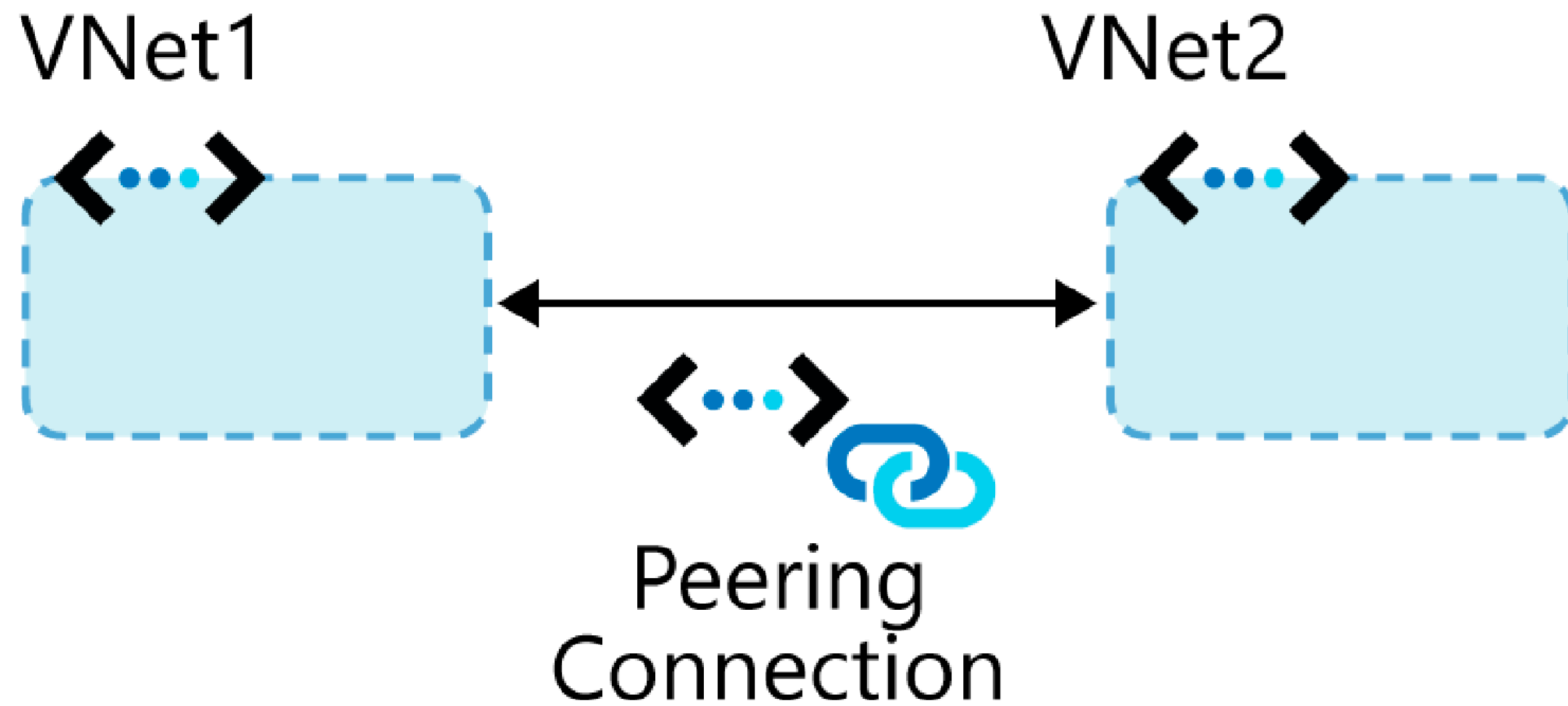
VPN Gateway
connections



ExpressRoute

Virtual Network Peering

It connects two Azure virtual networks.



Virtual Network Peering: Types

Virtual network peering

Connects virtual networks in the same Azure region

Example: Connecting two virtual networks in North Europe

Global virtual network peering

Connects virtual networks present in different Azure regions

Example: Connecting a virtual network in North Europe and a virtual network in West Europe

Peering Considerations

Reciprocal connections:

A user must establish a connection on each virtual network to link the networks while using virtual network peering.

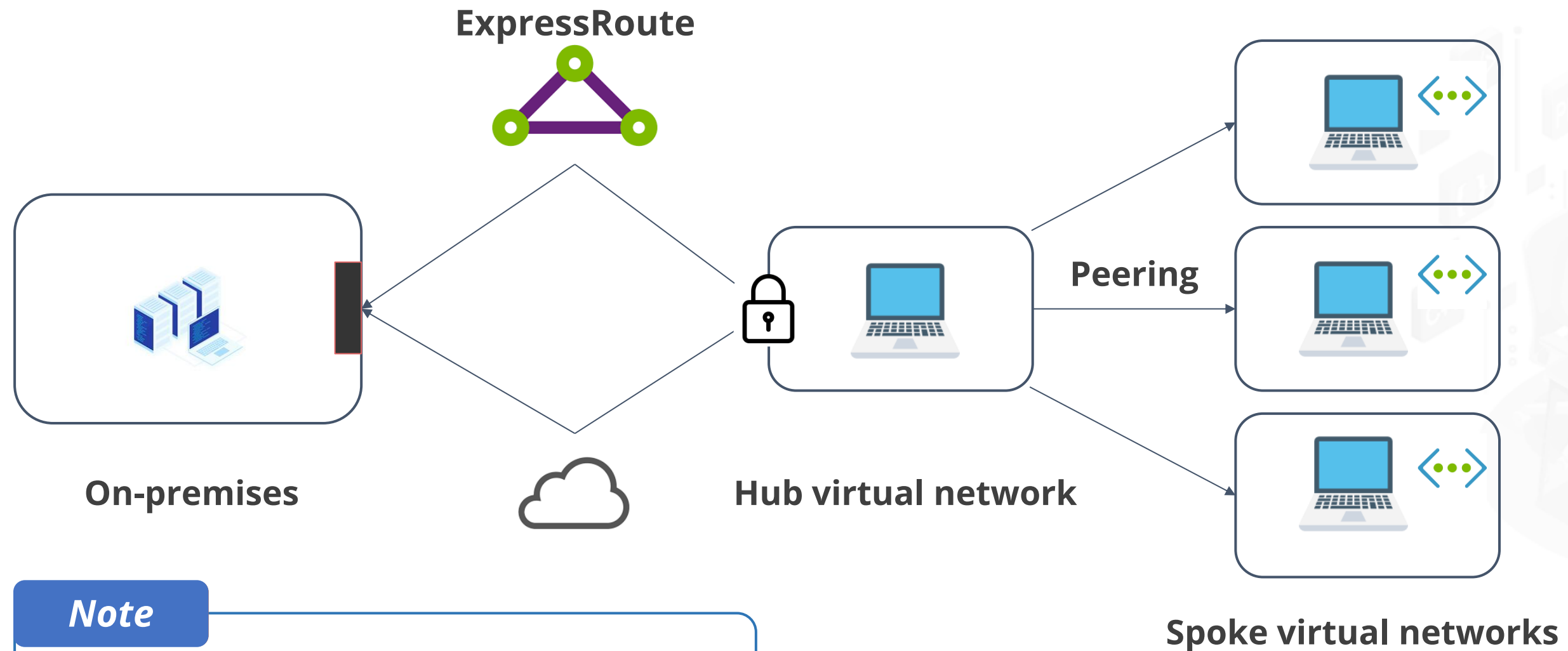
Cross-subscription connections:

Virtual network peering can be done even when both virtual networks are in different subscriptions.



Transitivity

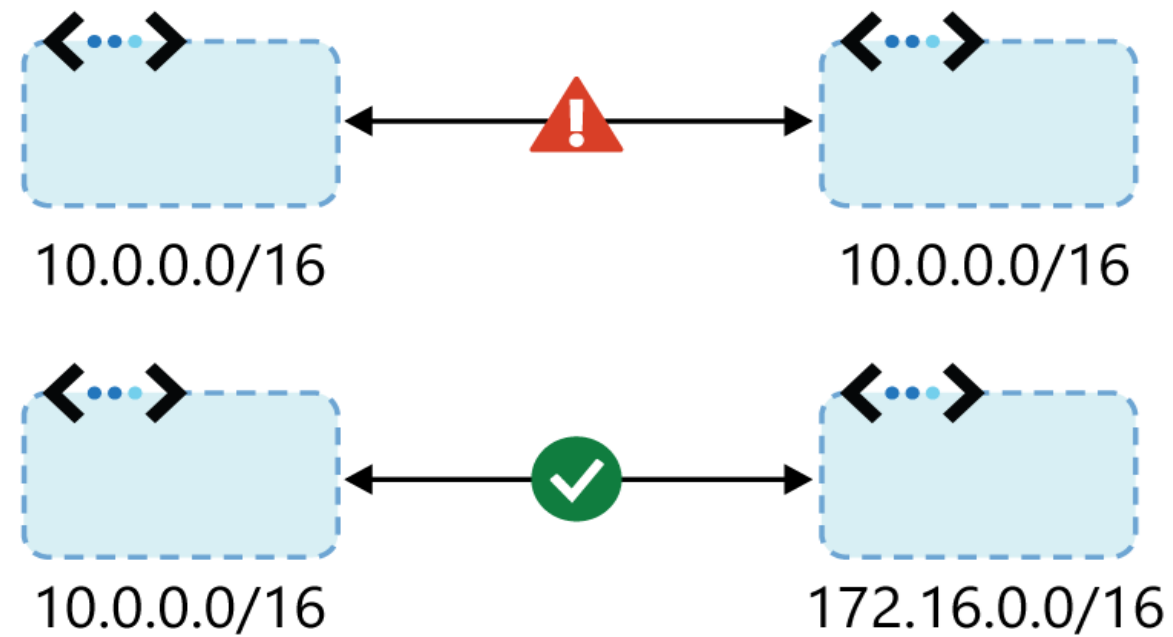
In Azure, peer-to-peer transitive routing involves network traffic between two virtual networks routed through an intermediate network.



Note

Virtual network peering is nontransitive.

Gateway Transit



Benefits

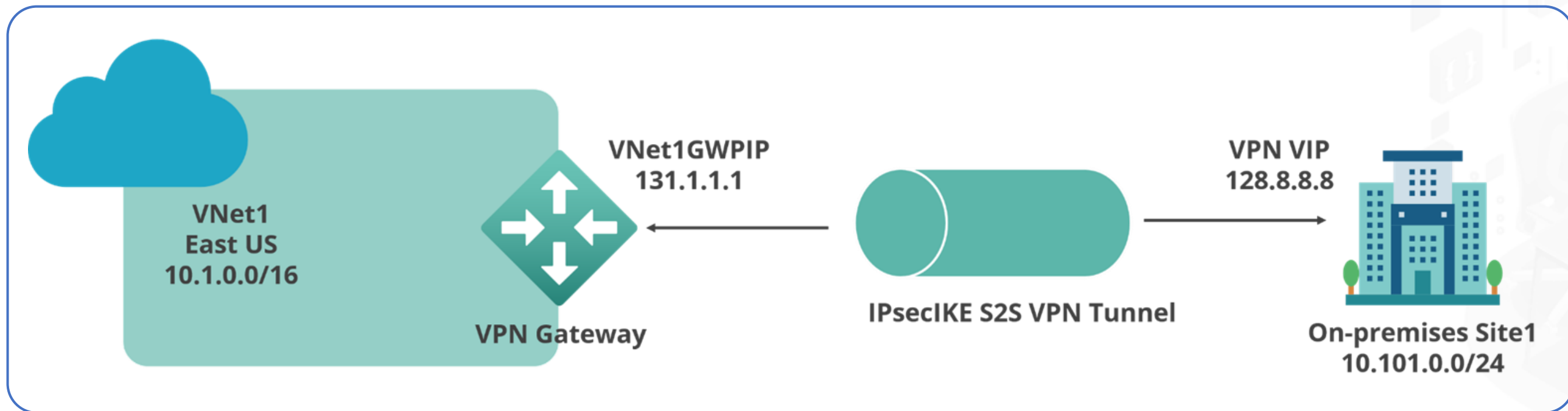
- Facilitate cross-premise connectivity
- Enable the allow gateway transit option in the hub
- Enable the use remote gateway option on the spoke

Peering considerations

- IP address spaces should not overlap
- Peering is the recommended option

VPN Gateways

It provides cross-premises connectivity between customer premises and Azure.



VPN Gateway Types



Site-to-Site

Connects on-premises data centers to Azure virtual networks



VNet-to-VNet

Connects Azure virtual networks (custom)



Point-to-Site

Connects individual devices to Azure virtual networks

VNet Peering vs. VPN Gateway vs. Gateway Transit

VNet peering

- Is direct (no interconnecting device)
- Has low latency and high bandwidth
- Is regional or global

VPN gateway

- Serves as an interconnecting device
- Introduces extra latency and limits bandwidth

Gateway transit

- Allows sharing a VPN or ExpressRoute gateway across a peering
- Minimizes complexity and centralizes management

VNet Peering vs. VPN Gateway

Item	Virtual network peering	VPN gateway
Limits	Up to 500 per VNet	One per VNet (per gateway limits are SKU-dependent)
Pricing model	Ingress/Egress	Hourly + Egress
Encryption	Not included	IPsec/IKE
Bandwidth	No limits	SKU-dependent
Public endpoints	No	Yes
Transitivity	No	Yes (routing-dependent)
Initial setup time	Fast	30 minutes
Typical scenarios	Data replication, database failover, and other scenarios needing frequent backups of large data	Encryption-specific scenarios are not latency-sensitive and do not need high power

VNet Peering and VPN Gateway

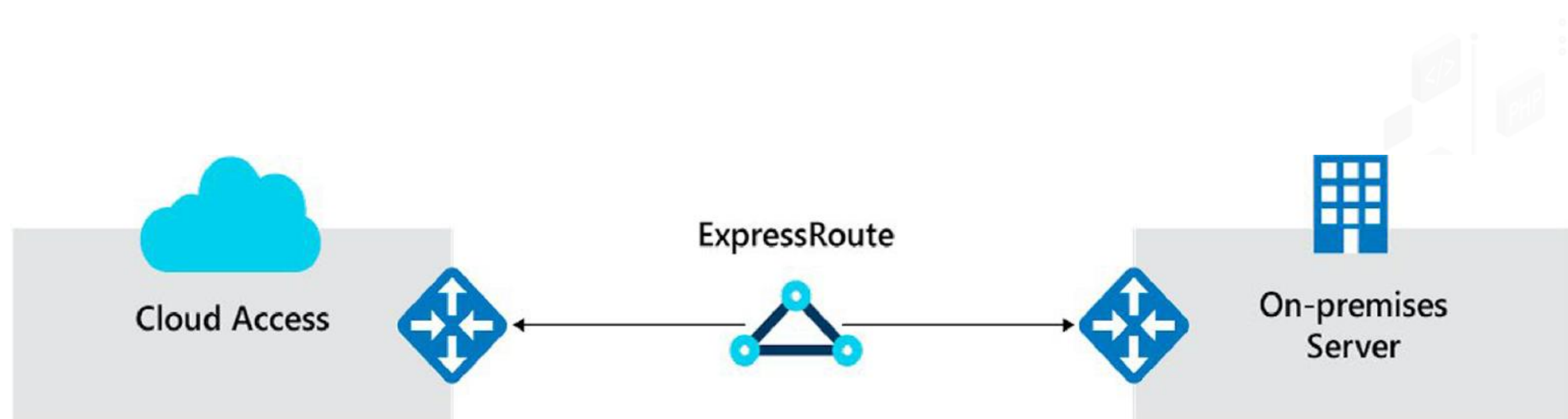


They facilitate connections between:

- Virtual networks in different regions
- Virtual networks in different Azure Active Directory tenants
- Virtual networks in different Azure subscriptions
- Virtual networks that use a mix of Azure deployment models

Azure ExpressRoute for Hybrid Networks

It is an Azure service that allows the user to extend on-premises networks over a private connection with the help of a network.



ExpressRoute Circuits

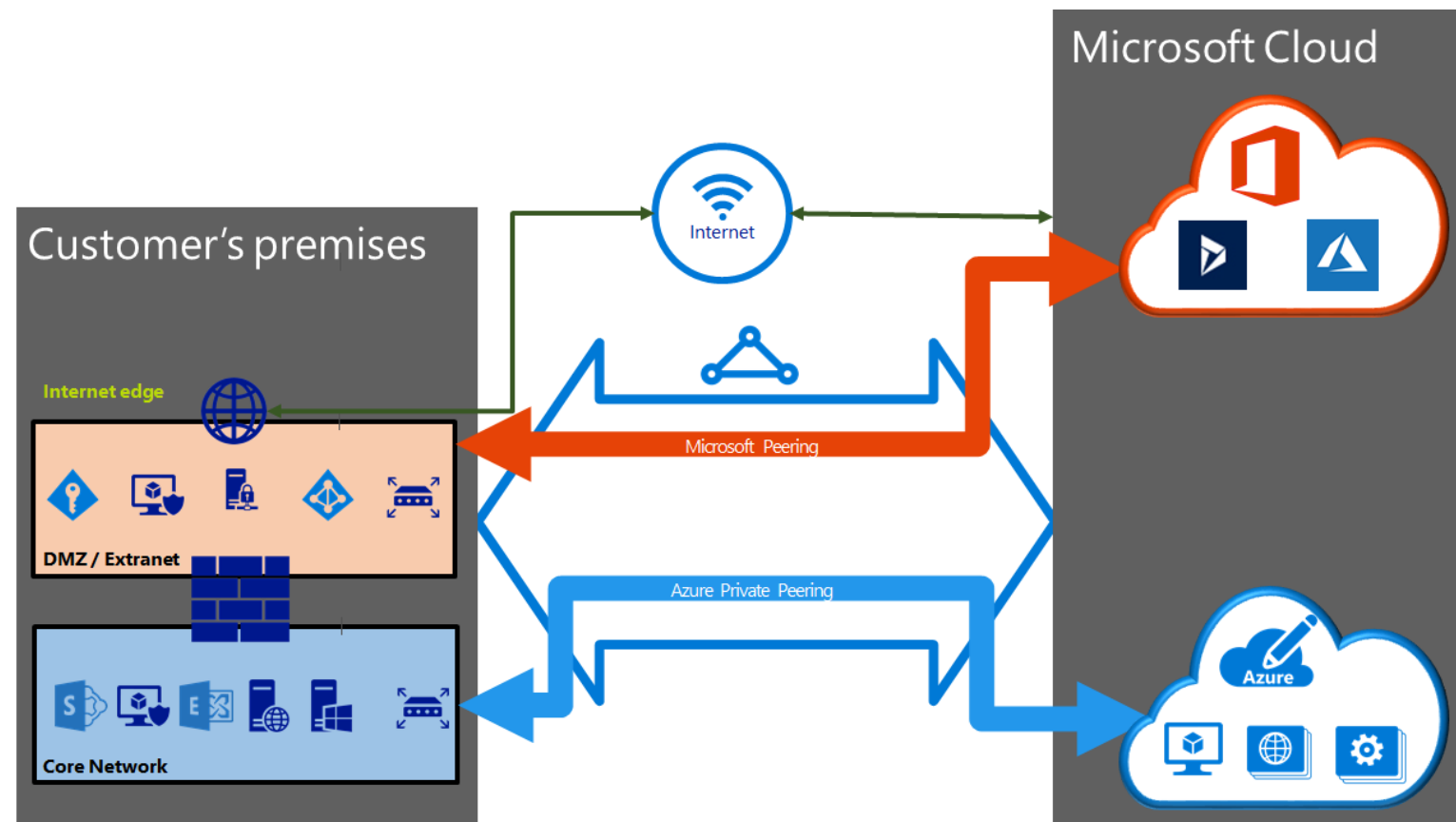
A Circuit is an ExpressRoute logical connection between an on-premises network and an Azure network.

Benefits

- Sets up traffic management and routing in ExpressRoute through circuits
- Enables the utilization of multiple circuits spanning different regions
- Facilitates connections through diverse connectivity providers
- Accommodates multiple routing domains and peering
- Operates without the requirement for physical mapping



ExpressRoute Peering



Azure private peering

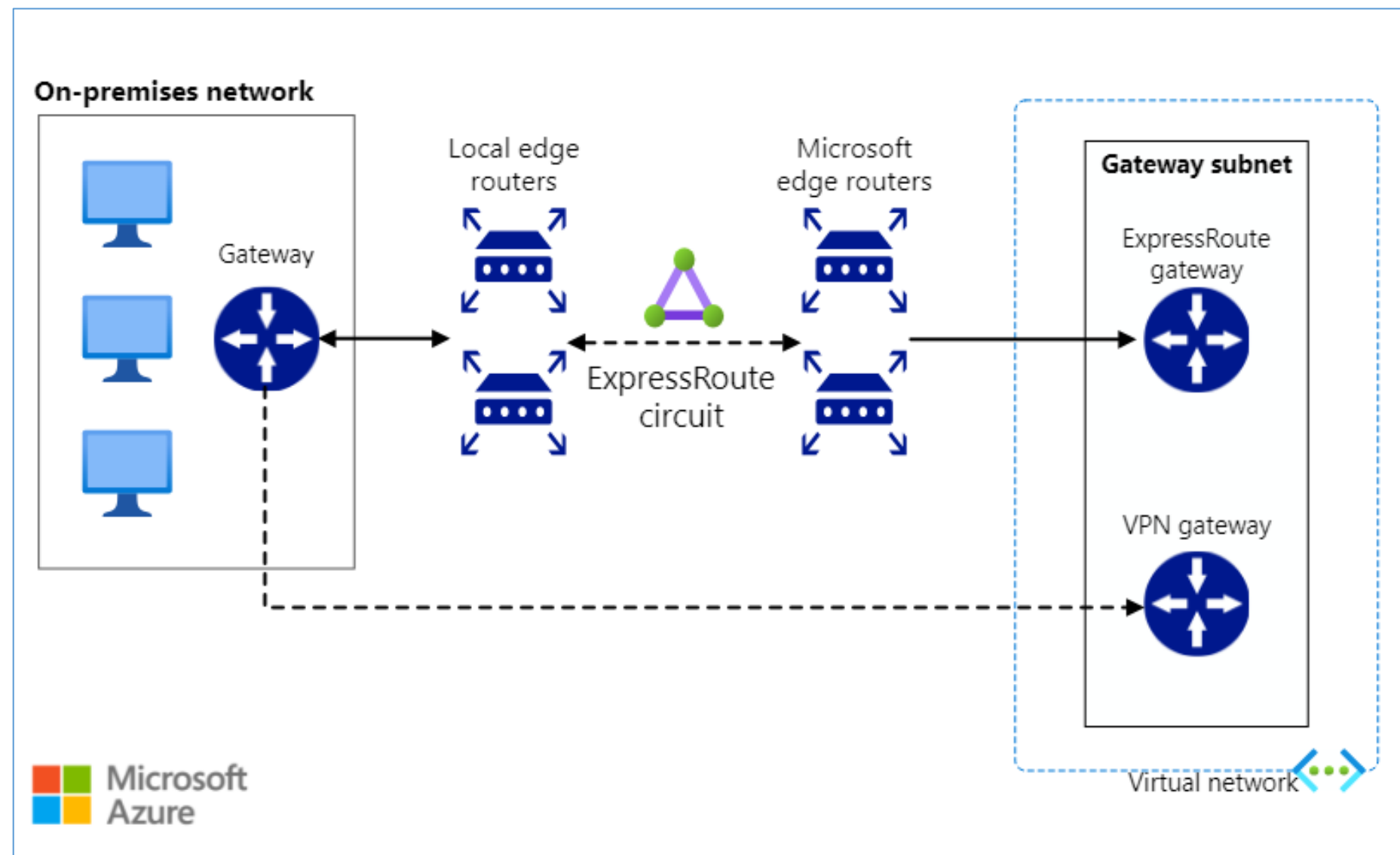
Connects to virtual machines and cloud services directly using private IP addresses

Microsoft peering

Provides connectivity to all Microsoft online services

ExpressRoute Reference Architecture

It shows how to connect an on-premises network to an Azure virtual network by using ExpressRoute, with a site-to-site VPN as a failover connection.



Source: <https://learn.microsoft.com/en-us/azure/architecture/reference-architectures/hybrid-networking/expressroute-vpn-failover>

ExpressRoute Considerations

1

Supports circuit management by the connectivity provider

2

Requires a Border Gateway Protocol (BGP) configuration

3

Requires working with connectivity providers

4

Requires on-premises high-bandwidth routers

5

Monitors the connectivity using the Azure Connectivity Toolkit

6

Requires network security appliances between the routers and networks

7

Requires a network security appliance to manage threats

Service Endpoint

It allows secure and direct access to Azure services via an efficient path across the Azure backbone network.



Endpoints enable the user to restrict access of Azure service resources exclusively to their virtual network.



Service Endpoint: Benefits

Benefits	Explanation
Improved security for Azure service resources	By extending the VNet identity to Azure service endpoints, users may secure Azure service resources to user's virtual network.
Optimal routing for Azure service traffic from the virtual network	Service traffic is always routed directly from the user's virtual network to the service on the Microsoft Azure backbone network by endpoints.
Simple to set up with less management overhead	Service endpoints do not require any Network Address Translation (NAT) or gateway devices to be set up.

Private Link

It allows a user to use a private endpoint in user virtual network to access Azure PaaS services as well as Azure hosted customer-owned or partner services.



Azure Private Link's setup and consumption are consistent across Azure PaaS, customer-owned, and shared partner services.

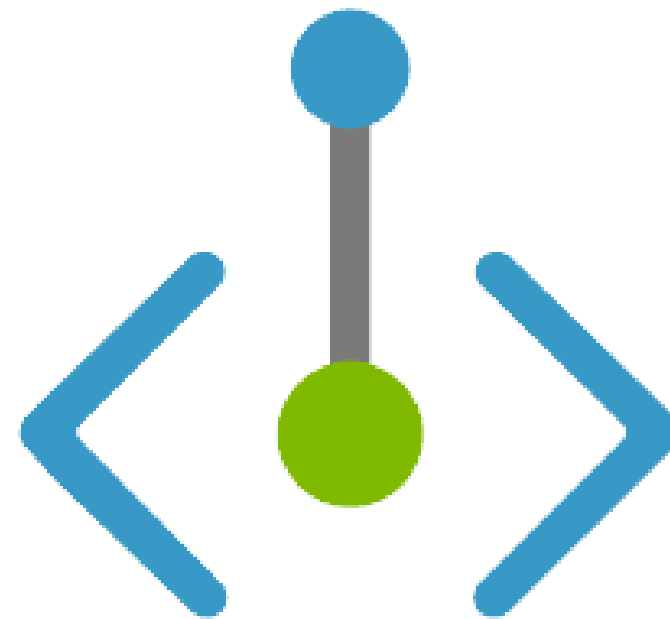


Private Link Benefits

Benefits	Explanation
Privately access services on Azure platform	Azure backbone network is used to link the customer to the services, which will be handled by the Private Link Platform.
On-premises and peered networks	Users must access Azure services from on-premises using private endpoints, ExpressRoute private peering, VPN tunnels, and peered virtual networks.
Protection against data leakage	A private endpoint is associated with a PaaS resource instance rather than the entire service.
Global reach	It helps to connect to services securely in other parts of the world.

Private Endpoint

It is a network interface that connects to a user's virtual network using a private IP address.



This network interface connects users to a service powered by Azure Private Link in a private and secured manner.



Network Security of Private Endpoints: Features

- Secures traffic to a private link resource when private endpoints are used
- Performs access control to ensure that network connections only go to the private link resource defined
- Prevents workloads from connecting to a supported Azure service via public endpoints
- Prohibits access to other Azure resources that are hosted on the same Azure service

Web Application Firewalls (WAFs)

It provides online applications with a basic level of protection and additional defense-in-depth mitigation.

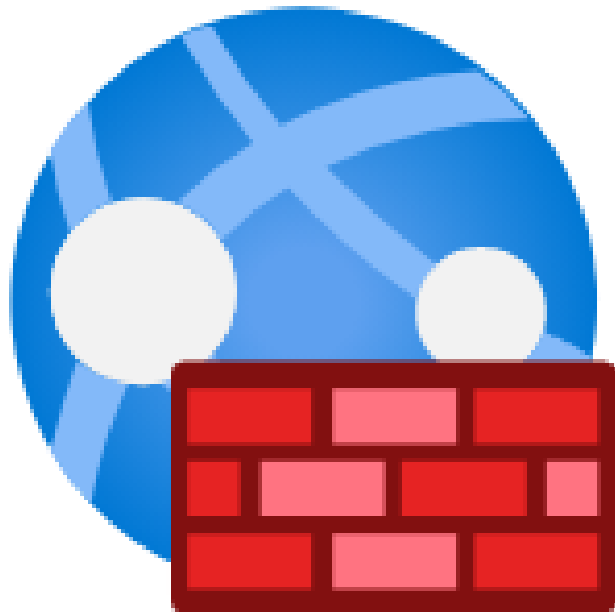


It helps to reduce the chances of an attacker exploiting common application security flaws.



Web Application Firewalls (WAFs)

This mechanism is an important mitigation, as attackers often target web applications as an entry point into an organization.



- WAF capabilities within Azure Application Gateway involve the inspection of web traffic and identification of threats at the HTTP layer.
- It functions as both a load balancer and a complete HTTP(S) reverse proxy, managing SSL traffic by encrypting and decrypting data.

Azure Firewall

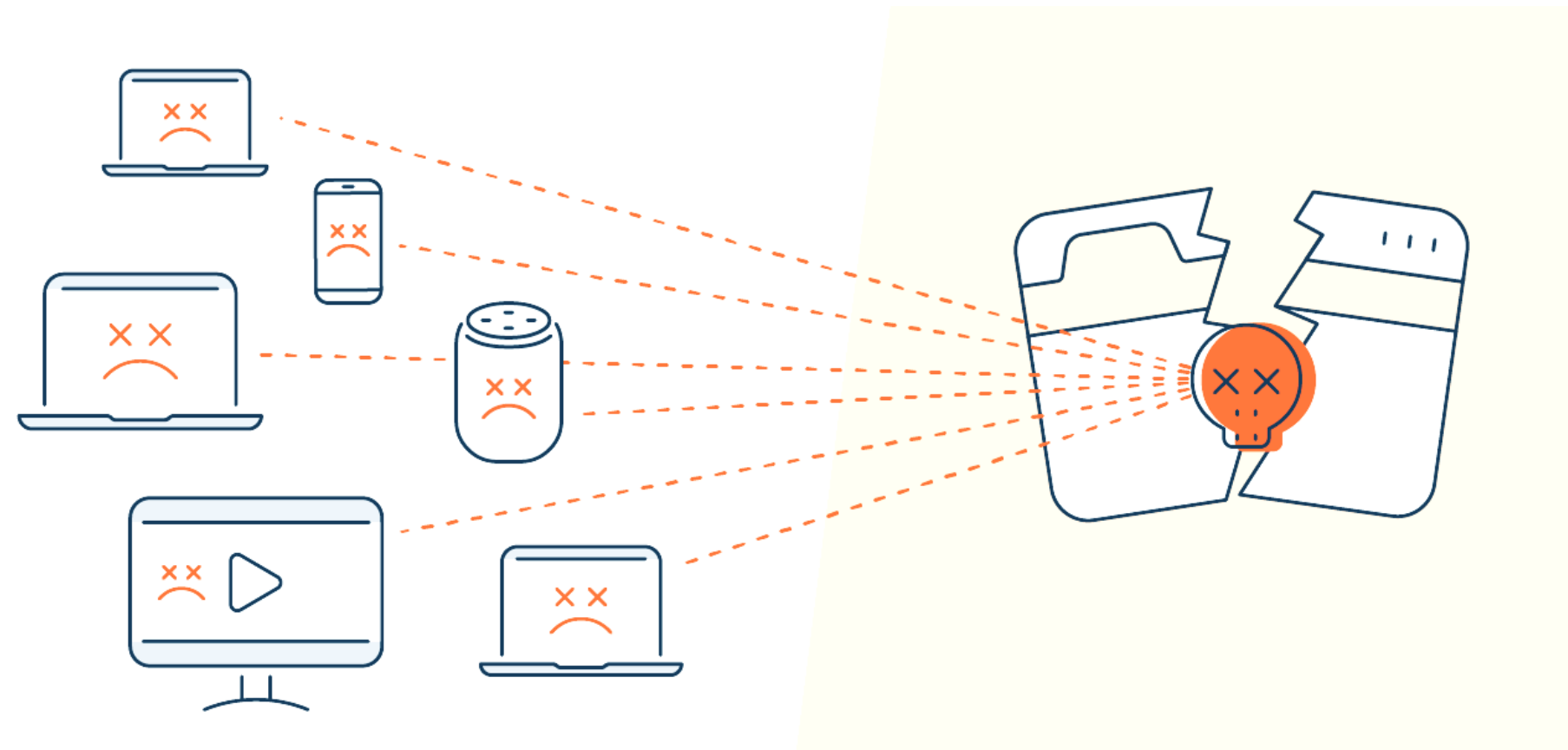
It is a cloud-native, intelligent network firewall security service that offers the best threat prevention for user Azure workloads.



- It protects the virtual network against potentially harmful traffic from the internet and other sources.
- It inspects incoming traffic and allows only authorized requests to proceed.

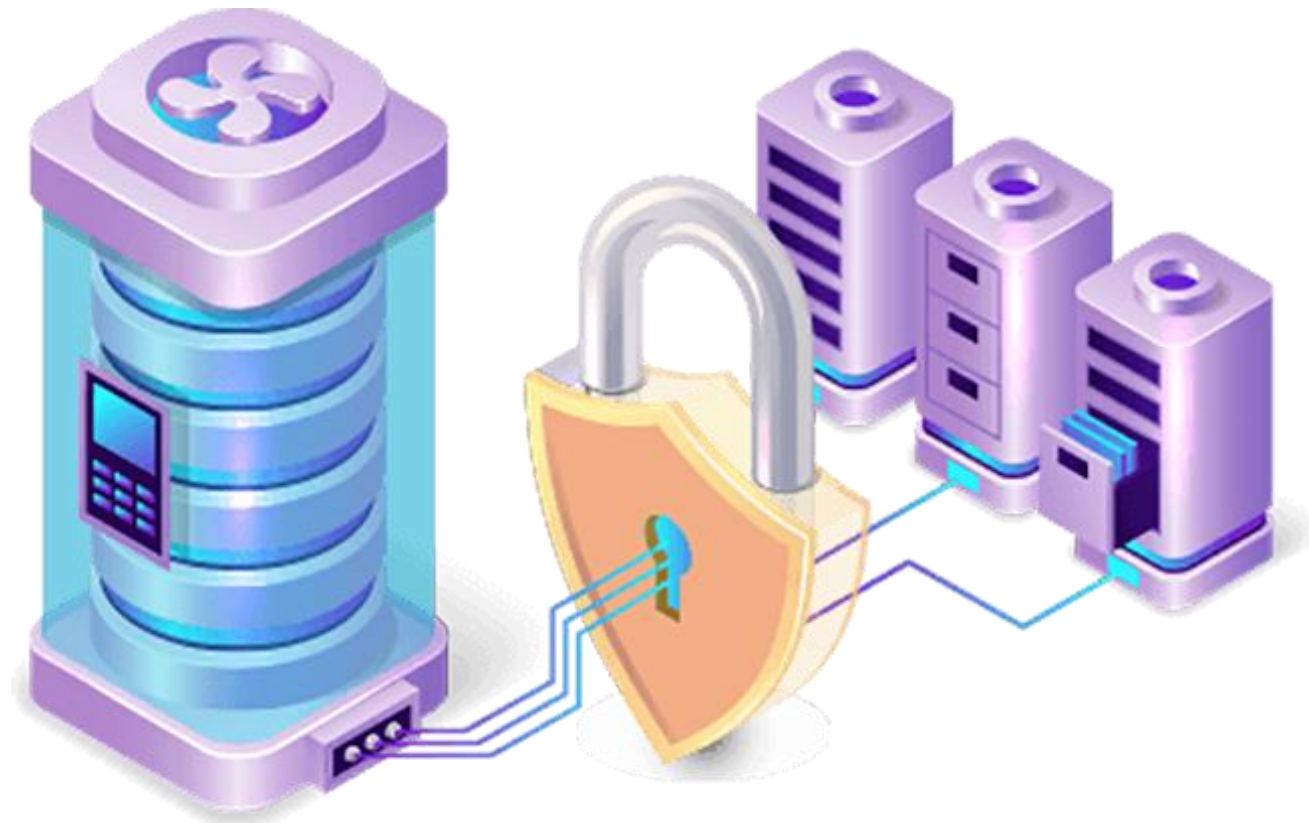
DDoS Attacks

In a Distributed Denial-of-Service (DDoS) attack, the server is overloaded with bogus traffic.



This attack can completely shut down or block access to services.

DDoS Protection



- DDoS assaults are already protected by Azure Infrastructure.
- For services provisioned via a virtual network, Azure provides additional security in the network.
- Features such as machine learning-based monitoring approaches must be considered for detecting unusual traffic.
- User application must be protected before it suffers a service degradation.

Azure Bastion

It is provisioned as a fully managed PaaS service to securely connect to virtual machines via a private IP address.

- Allows secure and smooth RDP and SSH communication to the user's virtual network's VMs through TLS, directly from the Azure portal
- Protects virtual machines from exposing RDP/SSH ports to the outside world, while still providing secure access using RDP/SSH

Just-in-Time (JIT) Access

JIT VM access is a Defender for Cloud feature that gives network-based access to VMs on time.



This feature secures the VM network interface or the subnet that contains the VM network interface by adding a deny rule to the Azure NSG.

Creating Virtual Networking and Resolving Domain Names



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you have been asked to create a virtual network in Azure that will host virtual machines. Also, you need to deploy them into different subnets of the virtual network, and you need to implement DNS name resolution for Azure virtual machines within the virtual network.

Outcome:

You will be able to set up a virtual network in Azure, deploy VMs across different subnets, and enable DNS name resolution for Azure virtual machines within this network.

Note: Refer to the demo document for detailed steps:
[01_Creating_Virtual_Networking_and_Resolving_Domain_Names](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Log in to the Azure portal
2. Create a virtual network
3. Create a subnet
4. Create a public IP address and the network interface
5. Create the network security group



Recommend Solutions for Automating Network Management

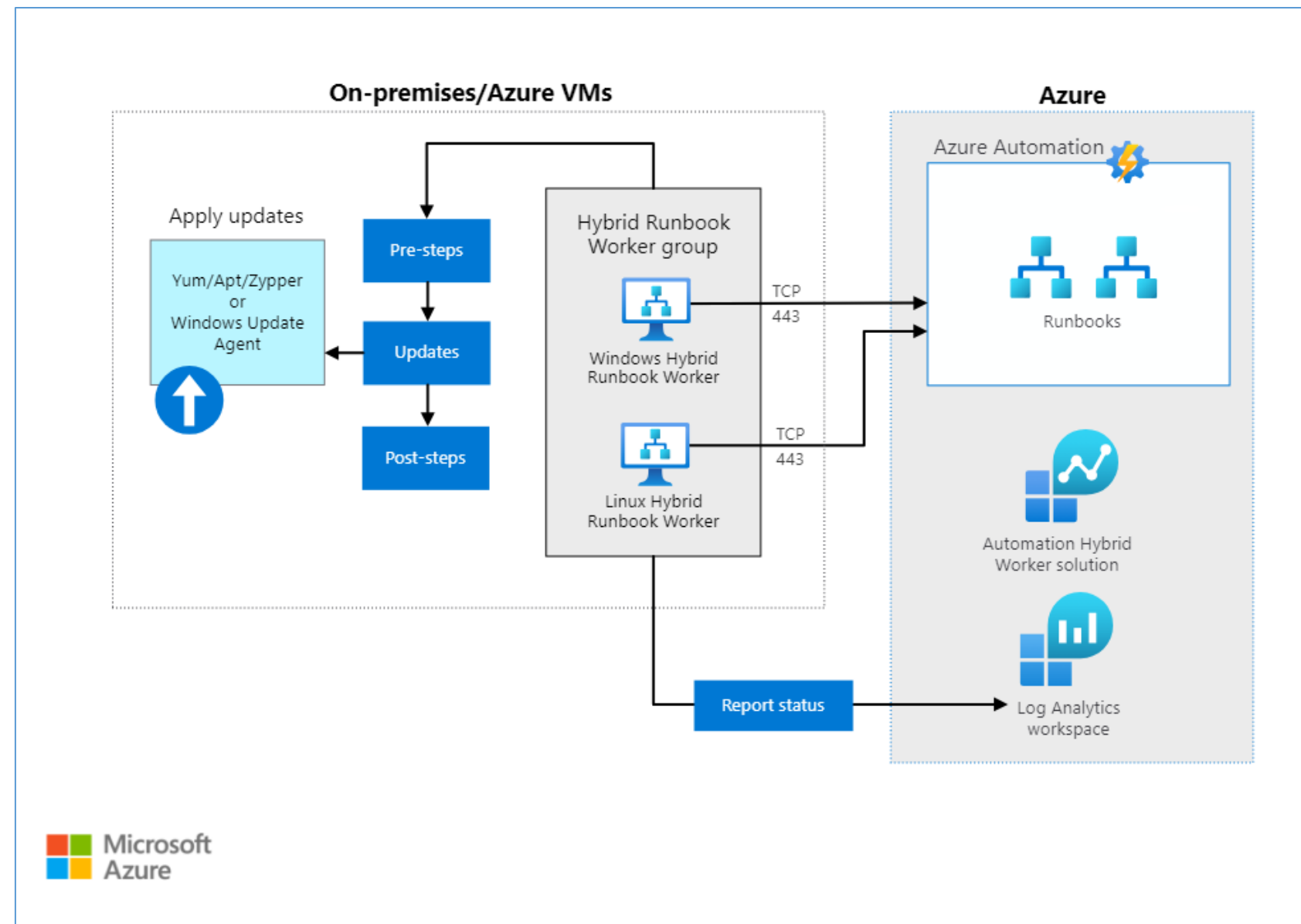
Azure Automation Update Management

It allows the user to handle operating system upgrades for virtual machines in Azure, on-premises, and other cloud environments.



Hybrid Update Management Architecture

This reference architecture illustrates how to design a hybrid update management solution to manage updates on both Microsoft Azure and on-premises Windows and Linux computers.



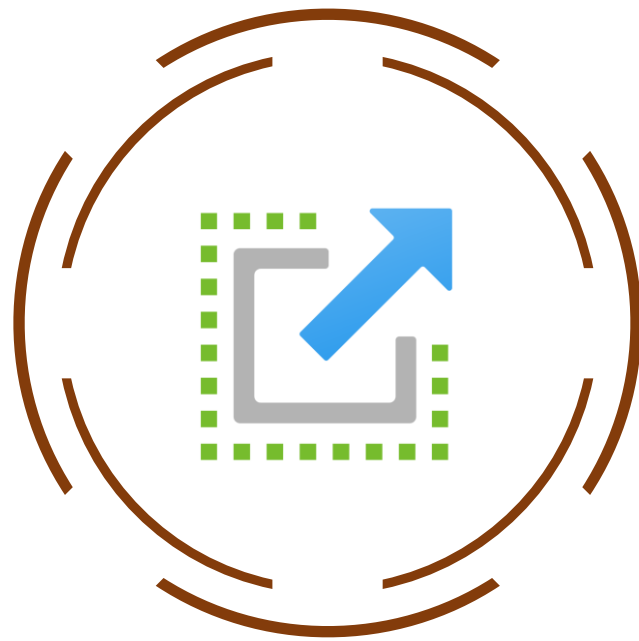
Source: <https://learn.microsoft.com/en-us/azure/architecture/hybrid/azure-update-mgmt>

Hybrid Update Management Architecture: Use Cases

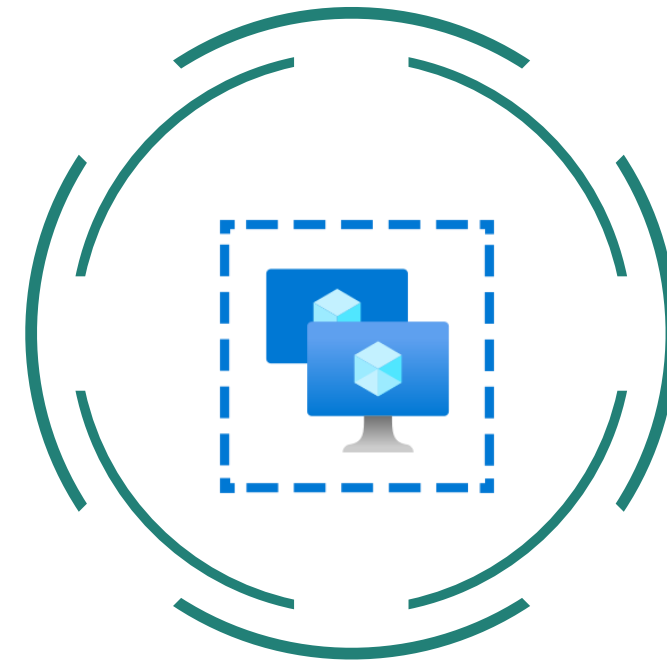
Managing updates across on-premises and in Azure using the Update Management component of the Automation Account

Using scheduled deployments to orchestrate the installation of updates within a defined maintenance window

Update Management Considerations



Scalability
considerations



Availability
considerations



Recommend Solutions for Load Balancing and Traffic Routing

Load Balancing and Traffic Routing

Azure offers these services for managing network traffic distribution and load balancing:



Load
Balancers



Application
Gateway



Front
Doors

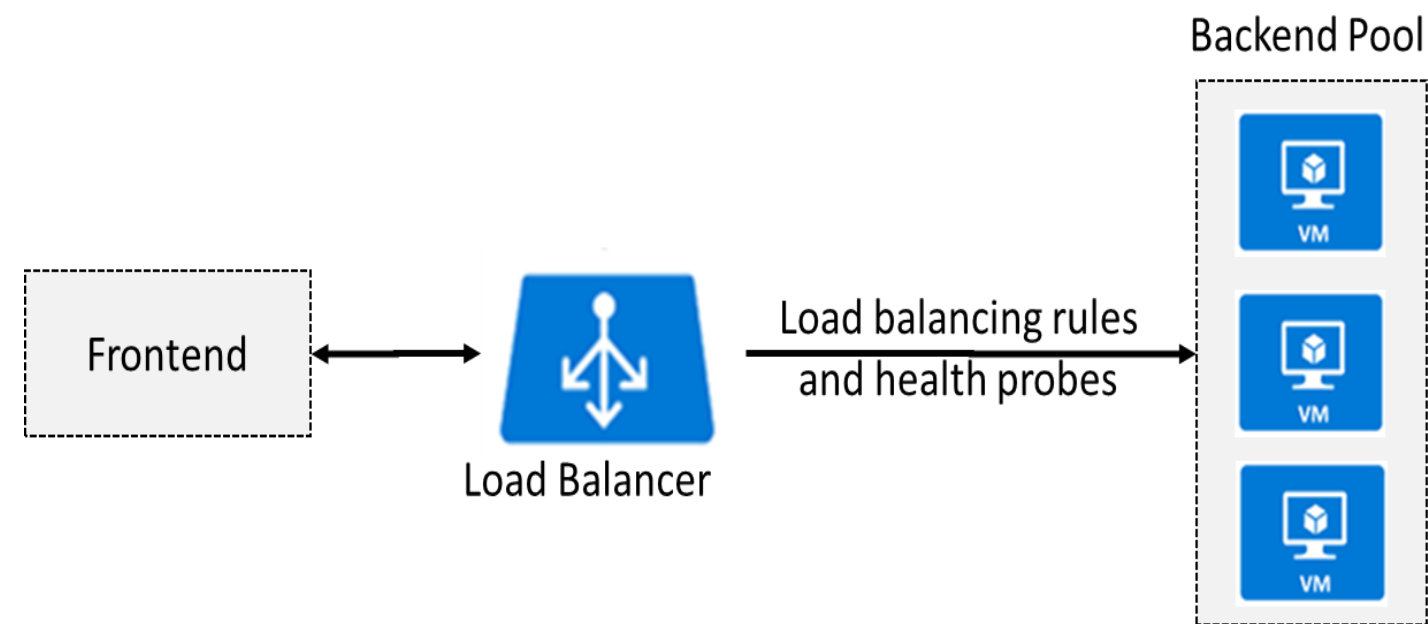


Traffic
Manager

Based on the user's demands, they can use these services individually or in combination to create the best solution.

Azure Load Balancer

It distributes inbound traffic to backend resources using load balancing and health probes.

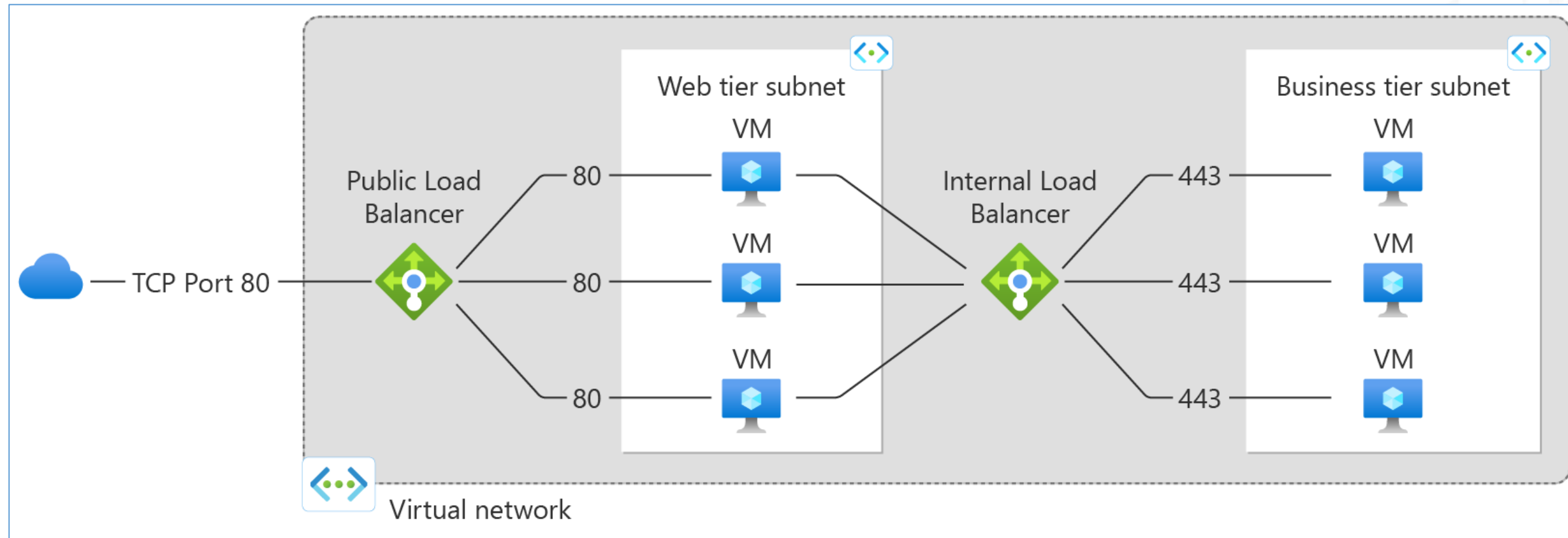


The on-premises web applications are protected with secure remote access.

Azure Load Balancer: Types

Public Load Balancer

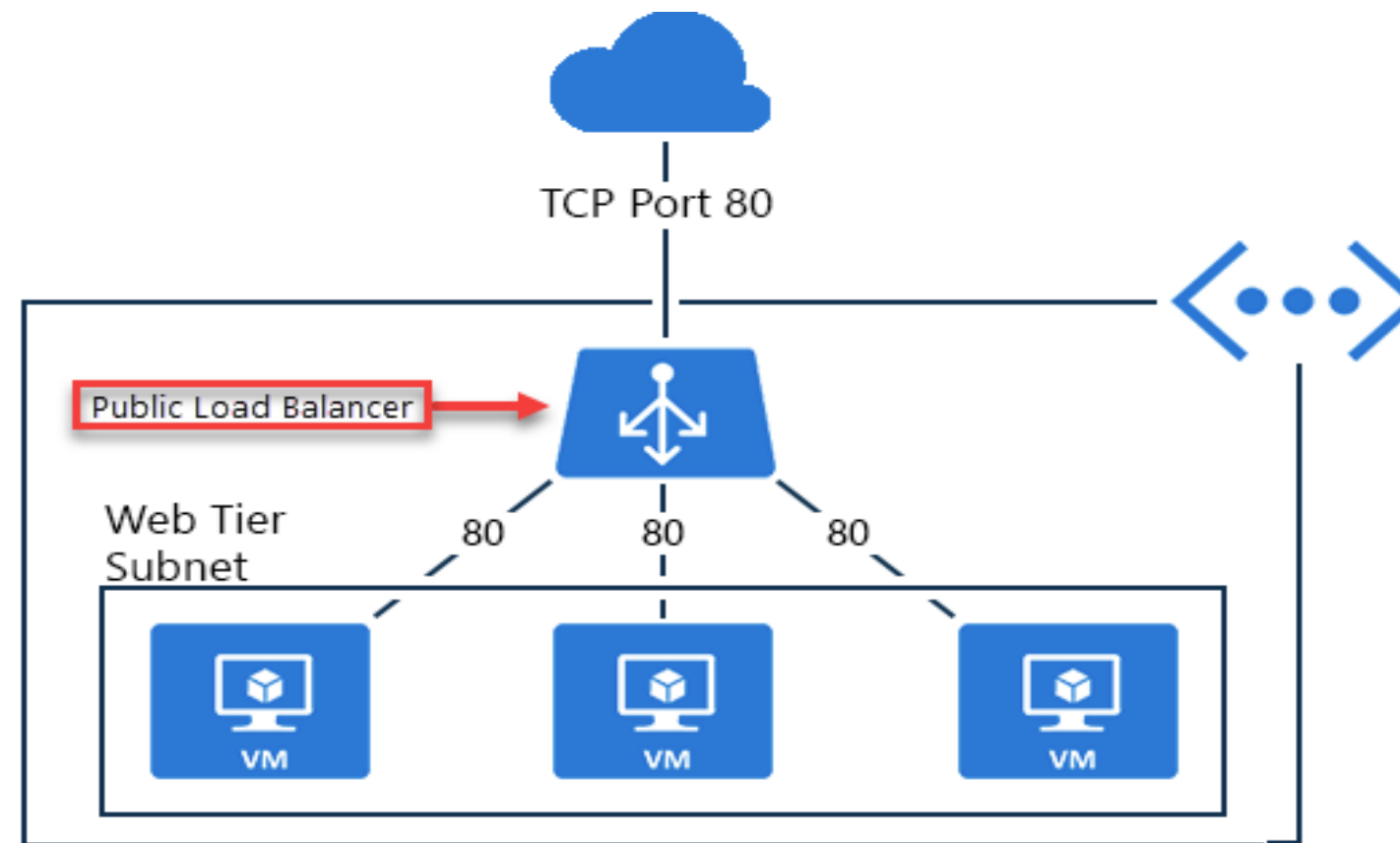
Internal Load Balancer



Source: <https://learn.microsoft.com/en-us/azure/load-balancer/load-balancer-overview>

Public Load Balancer

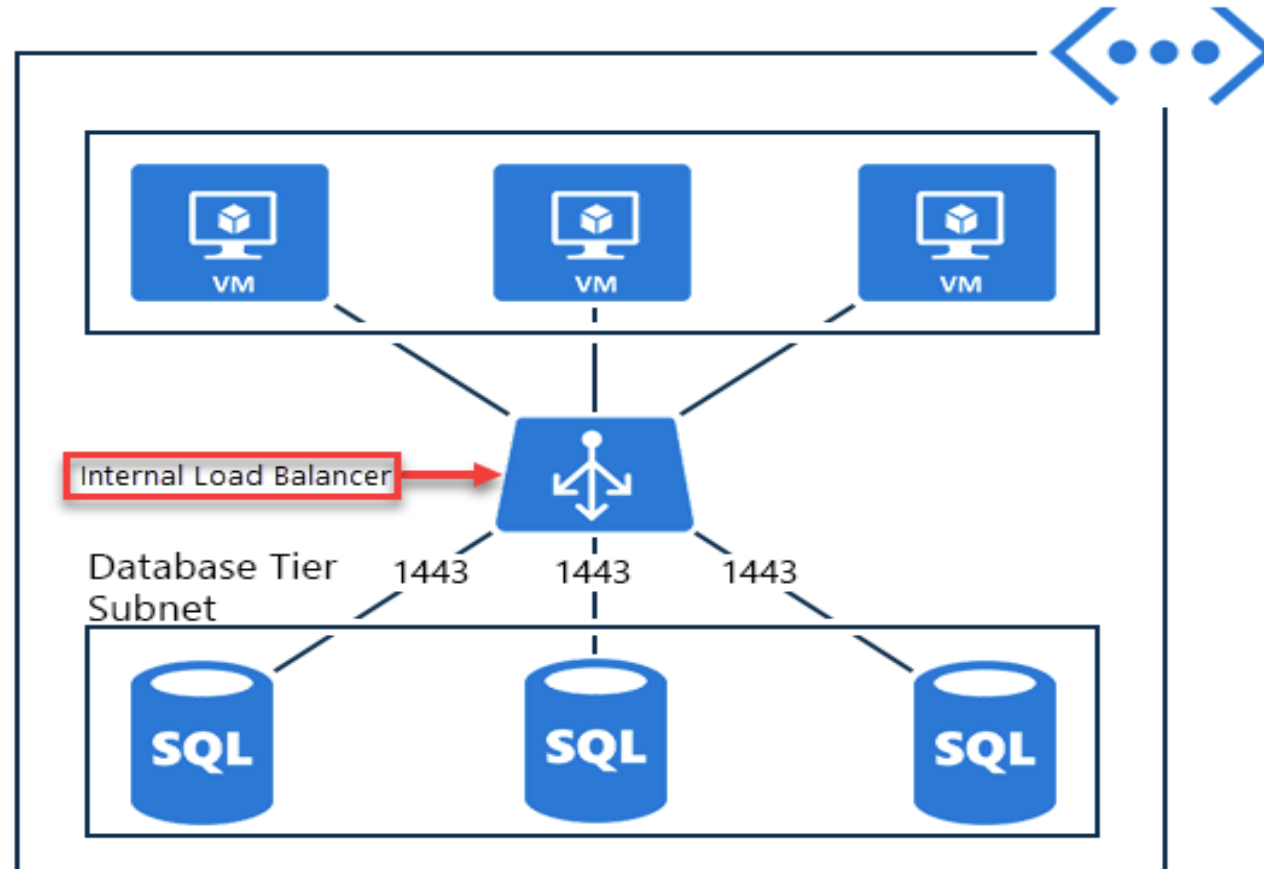
It maps public IP addresses and port numbers of incoming traffic to the VM's private IP address and port number and vice versa.



It applies load balancing rules to distribute traffic across VMs or services.

Internal Load Balancer

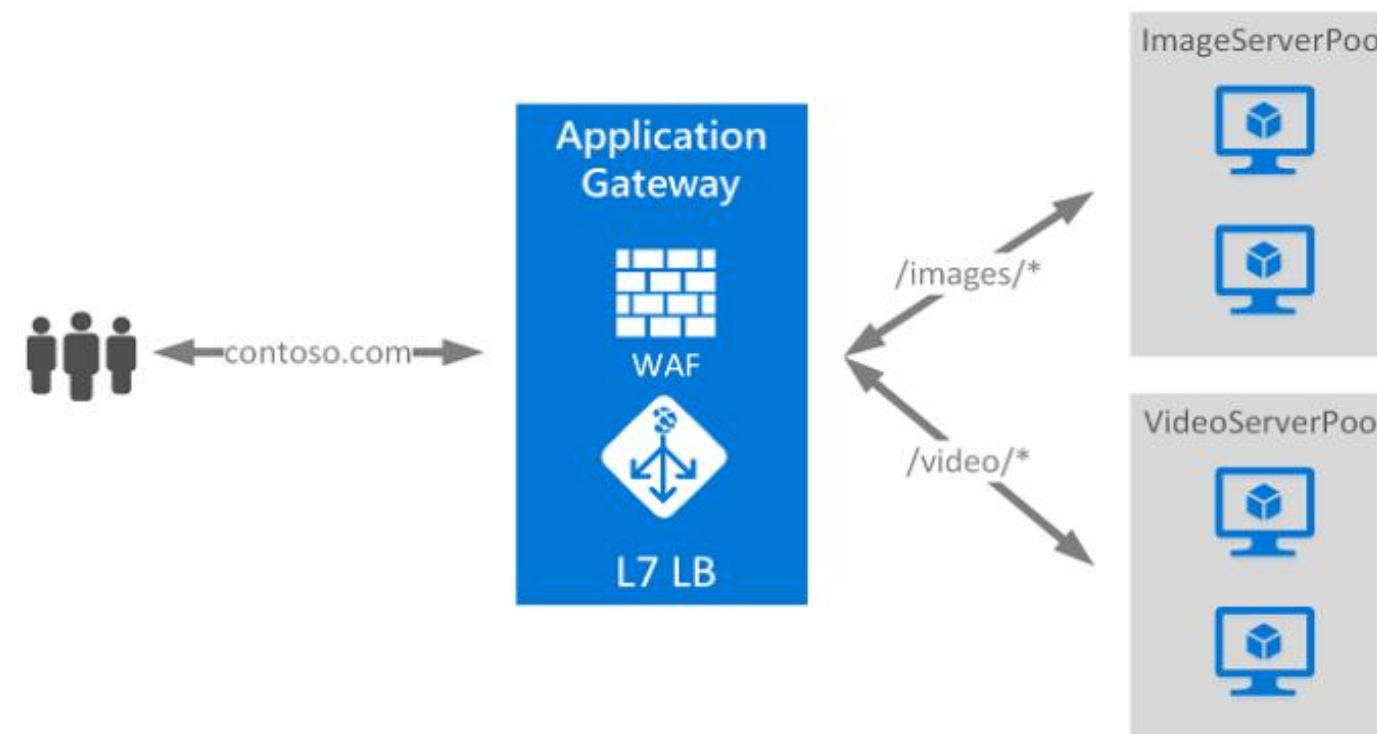
It directs traffic only to resources inside a virtual network or those that use a VPN to access Azure infrastructure.



It enables load balancing within a virtual network for cross-premises virtual networks, multi-tier applications, and line-of-business applications.

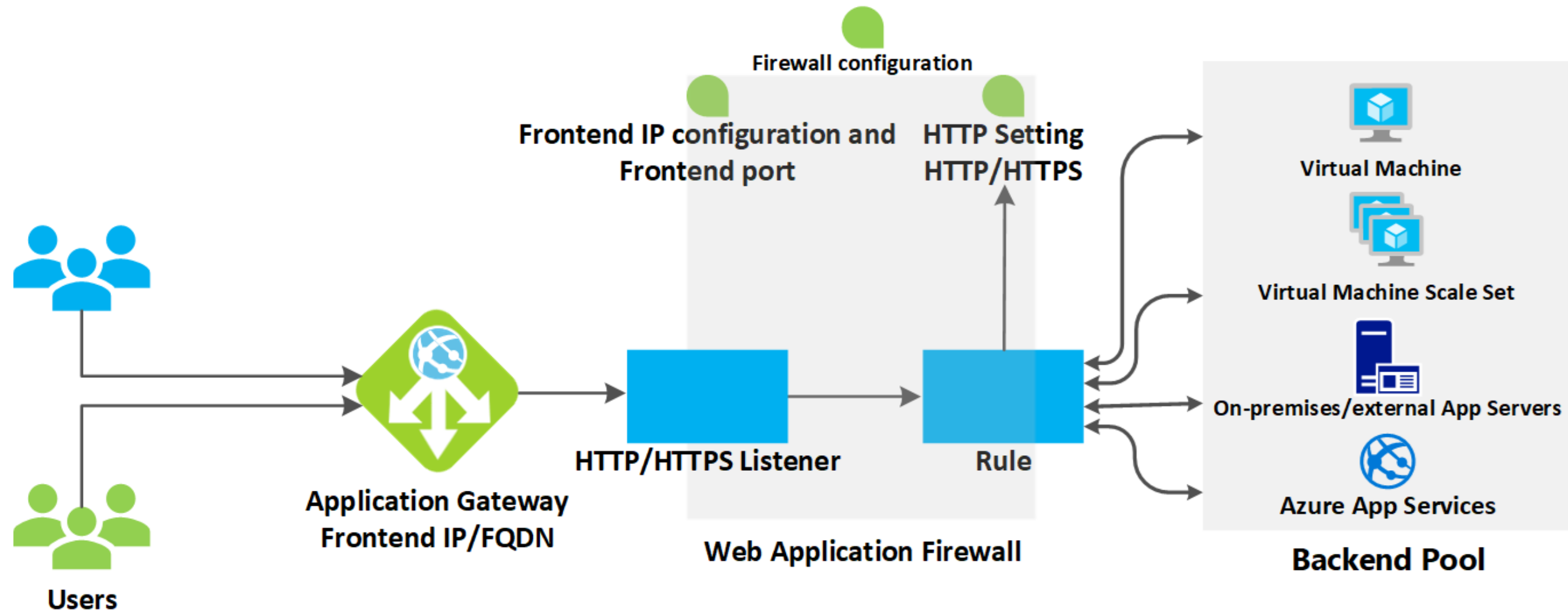
Application Gateway

It manages the requests that client applications can send to a web application.



It routes traffic to a pool of web servers based on the URL of a request.

Application Gateway Workflow



Source: <https://learn.microsoft.com/en-us/azure/application-gateway/how-application-gateway-works>

Traffic Manager

It enables the distribution of traffic to public-facing applications across Azure regions globally.

Offers public endpoints with high availability and quick responsiveness

Implements health monitoring for every endpoint

Provides a range of traffic-routing methods and endpoint monitoring



Traffic Manager: Features

Increases application availability

Improves application performance

Distributes traffic for complex deployments

Maintains service without downtime

Combines hybrid applications



Distributing Network Traffic

Service	Azure Load Balancer	Application Gateway	Traffic Manager	Azure Front Door
Technology	Transport layer (Level 4)	Transport layer (Layer 7)	DNS resolver	Layer 7 or HTTP/HTTPS
Protocols	Any TCP or UDP protocol	HTTP, HTTPS, HTTP/2, and web sockets	DNS resolution	Split TCP-based anycast protocol
Backends and endpoints	Azure VMs and Azure VM scale sets	Azure VMs, Azure VM scale sets, Azure app services, IP addresses, and hostnames	Azure cloud services, Azure app services, Azure app services slots, and public IP addresses	Internet-facing services hosted inside or outside of Azure
Network connectivity	External and Internal	External and Internal	External	External and Internal

Creating Load Balancer



Duration: 10 Min.

Problem Statement:

You have been asked to assist your organization with an Azure network solution that can be used to evenly distribute the load (incoming network traffic) across a collection of backend resources or servers as an Azure Architect. Your applications should be scalable, and you should be able to construct highly available services.

Outcome:

You will be able to set up a robust Azure load balancer to efficiently distribute incoming network traffic across multiple servers, ensuring optimal performance and reliability of your applications.

Note: Refer to the demo document for detailed steps:
Demo_02_Creating_Load_Balancer

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Log in to your Azure portal
2. Create a resource
3. Select the option of networking and click on the load balancer
4. Create the load balancer



Implementing Intersite Connectivity



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you need to implement a lab environment with different networks with full connectivity between them and verify its functionality.

Outcome:

You will be able to seamlessly integrate multiple networks in Azure to create a fully connected lab environment, ensuring comprehensive interoperability and testing of network functionality.

Note: Refer to the demo document for detailed steps:
Demo_03_Implementing_Intersite_Connectivity

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

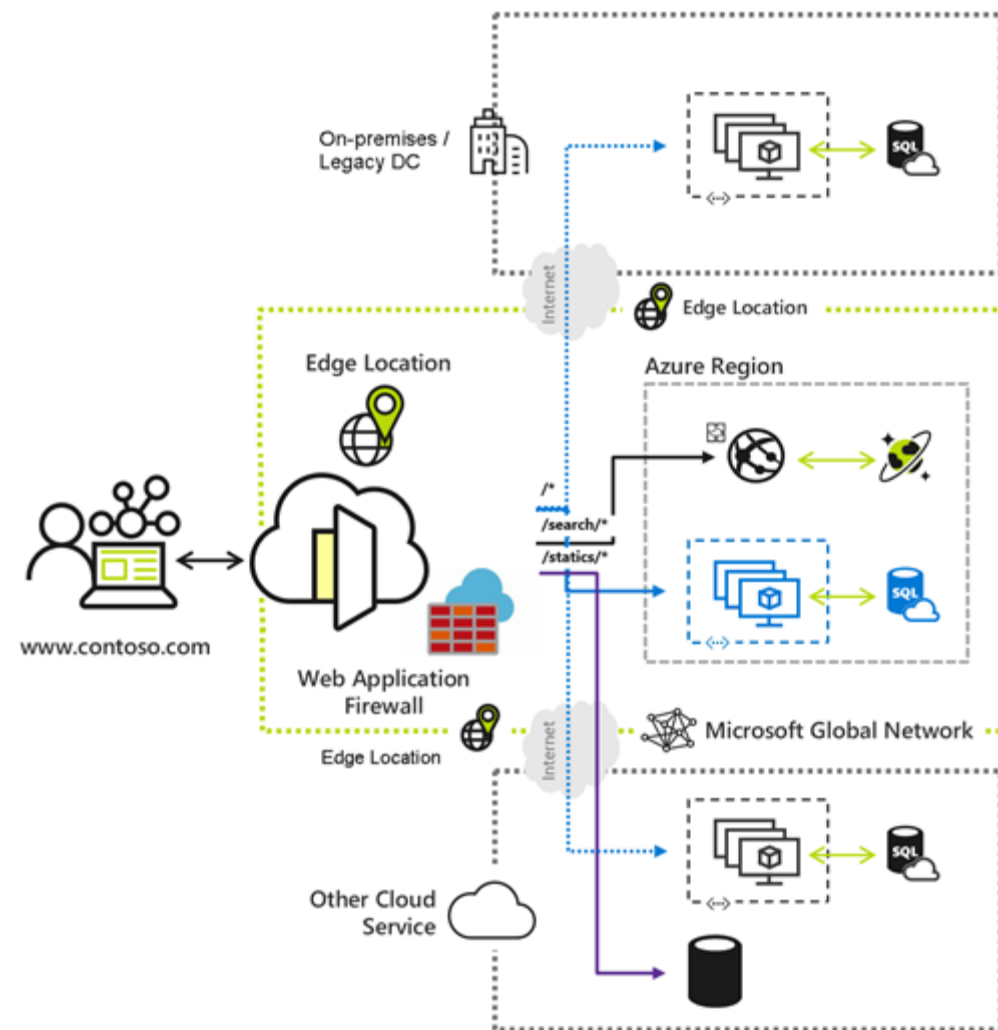
1. Provision the lab environment
2. Configure the local and global virtual network peering
3. Test intersite connectivity



Azure Front Door

Azure Front Door

This service is a global, scalable, and secure entry point that optimizes application delivery, providing intelligent routing, caching, and security features.



Source: <https://learn.microsoft.com/en-us/azure/frontdoor/front-door-overview>

Azure Front Door: Benefits

Ensure requests are directed to backends with the lowest latency

Implement primary and secondary backends for priority-based routing

Distribute traffic using weight coefficients for a balanced load

Integrate WAF and/or CDN for HTTP(s)-based traffic security and optimization

Maintain affinity by directing requests from the same end user to the same backend

Azure Front Door: Components

Frontend domains

Define custom domains for incoming traffic
Example: www.yourdomain.com

Backend pools

Group backend resources for load balancing
Example: Azure web apps, API Management

Health probes

Monitor backend health to ensure optimal traffic distribution
Example: Regularly check backend endpoints for responsiveness

Routing rules

Define routing logic based on path, host, or query parameters
Example: Route /api/* to API backend, /* to web frontend

Azure Front Door: Use Cases

Global application deployment

Scenario: Deploy applications globally with low-latency access

Use case: Global e-commerce platforms, content delivery networks

Enhanced security

Scenario: Strengthen application security with WAF capabilities

Use case: Protect against SQL injection, cross-site scripting (XSS)

Optimized performance

Scenario: Improve performance with SSL offloading and caching

Use case: Accelerate content delivery for web applications

Azure CDN (Content Delivery Network)

Azure CDN

It is a globally distributed network of servers that efficiently delivers web content, including images, videos, scripts, and stylesheets to users based on their geographical location.

Benefits

- Reduce latency and enhance the user experience by serving content from the nearest edge location
- Improve website and application load times
- Encrypt data in transit, ensuring a secure user experience
- Optimize delivery of dynamic content for faster page loads

Azure CDN: Use Cases

Global user base

Scenario: Serve content to users globally

Use case: International websites, global applications, media streaming

Improved performance

Scenario: Optimize website and application performance

Use case: Reduce load times, enhance user experience, and handle traffic spikes

Bandwidth cost reduction

Scenario: Minimize data transfer costs for content delivery

Use case: Reduce expenses associated with data transfer from origin servers

Azure CDN: Use Cases

Secure content delivery

Scenario: Ensure secure transmission of web content

Use case: E-commerce transactions, sensitive data delivery

Media streaming

Scenario: Efficiently deliver audio and video content

Use case: Live streaming, on-demand video, podcasts

Azure CDN: Features

Streams media and downloads large files quickly with optimized delivery

Provides a scalable solution to handle worldwide traffic spikes and instantaneous high loads

Integrates seamlessly with Azure services to activate within minutes

Protects content with custom domain HTTPS, DDoS, and WAF protection

Azure CDN: Best Practices

Optimize cache settings

Fine-tune caching rules for different types of content

Monitor and analyze
performance

Regularly analyze CDN performance metrics for
optimization opportunities

Utilize custom
domains

Enhance branding and user trust by using custom
domains with Azure CDN

Key Takeaways

- Application gateway manages requests and routes traffic to a pool of web servers.
- Traffic Manager routes users to web app endpoints in potentially different data centers located around the world.
- Network security protects the communication of resources within and outside of your network.
- NSG can be used to filter network traffic to and from Azure resources.
- Azure Front Door service is a global, scalable, and secure entry point that optimizes application delivery, providing intelligent routing, caching, and security features.





Knowledge Check

**Knowledge
Check
01**

Which Azure resource allows even splitting of incoming traffic and directing it to several identical virtual machines to perform the work and respond to requests?

- A. Virtual network
- B. Azure App Services
- C. Azure Load Balancer or Application Gateway
- D. Azure Logic Apps



**Knowledge
Check
01**

Which Azure resource allows even splitting of incoming traffic and directing it to several identical virtual machines to perform the work and respond to requests?

- A. Virtual network
- B. Azure App Services
- C. Azure Load Balancer or Application Gateway
- D. Azure Logic Apps



The correct answer is **C**

Azure Load Balancer or Application Gateway resource allows evenly splitting incoming traffic and directing it to several identical virtual machines to perform the work and respond to requests

**Knowledge
Check
02**

Which of the following services routes users to web app endpoints?

- A. Azure Front Door
- B. Azure Traffic Manager
- C. Backend Pool
- D. Application Gateway



**Knowledge
Check
02**

Which of the following services routes users to web app endpoints?

- A. Azure Front Door
- B. Azure Traffic Manager
- C. Backend Pool
- D. Application Gateway



The correct answer is **B**

Azure Traffic Manager routes users to web app endpoints.

**Knowledge
Check
03**

Which of the following options can be used to override the default network traffic rule? (Choose two)

- A. Azure Network Watcher
- B. Route Table
- C. Border Gateway Protocol Rules
- D. Application Gateway



Knowledge
Check
03

Which of the following options can be used to override the default network traffic rule? (Choose two)

- A. Azure Network Watcher
- B. Route Table
- C. Border Gateway Protocol Rules
- D. Application Gateway



The correct answer is **B and C**

Route Table and Border Gateway Protocol routes can be used to override the default network traffic rule.

**Knowledge
Check**
04

Which of these methods can be used when resources deployed in virtual networks need to resolve domain names to internal IP addresses?

- A. Azure DNS private zones
- B. Azure-provided name resolution
- C. Name resolution that uses your own DNS server
- D. All of the above



**Knowledge
Check**
04

Which of these methods can be used when resources deployed in virtual networks need to resolve domain names to internal IP addresses?

- A. Azure DNS private zones
- B. Azure-provided name resolution
- C. Name resolution that uses your own DNS server
- D. All of the above



The correct answer is **D**

When resources deployed in virtual networks need to resolve domain names to internal IP addresses, one of the three methods can be used.

**Knowledge
Check
05**

Which of the following is NOT a benefit of the hub-spoke architecture?

- A. Cost savings
- B. Workload isolation
- C. Limited subscription
- D. None of these



Knowledge
Check
05

Which of the following is NOT a benefit of the hub-spoke architecture?

- A. Cost savings
- B. Workload isolation
- C. Limited subscription
- D. None of these



The correct answer is **C**

Limited subscription is not a benefit of hub-spoke architecture.

Design a Network Solution

Duration: 25 min.



Project Agenda: To design a network solution for the below scenario

Objective: You are working as a principal engineer in an organization, and you have been asked to design a solution for an on-premise network to deploy a virtual appliance. The plan is to deploy multiple Azure VMs and connect the on-premises network to Azure using a site-to-site connection.

You also need to ensure that all the network traffic that will be directed from the Azure VMs to a specific subnet must flow through the virtual appliance. As part of compliance, the VMs should not access the internet as they store sensitive data. Recommend a design considering the above requirements.

Perform the following:

1. Logging in to the Azure portal
2. Creating a virtual network in the Azure portal
3. Deploying the virtual machine in the respective virtual network
4. Verifying the network security group of the virtual machine

TECHNOLOGY

Thank You